

Accessibility & Non-Visual Modalities

Alex Kale

Slides credited to Arvind Satyanarayan
& Catherine D'Ignazio

Medical (or deficit) model of disability: disability is something "wrong" with a person's body or mind (i.e., they are "suffering" and need to be "cured").

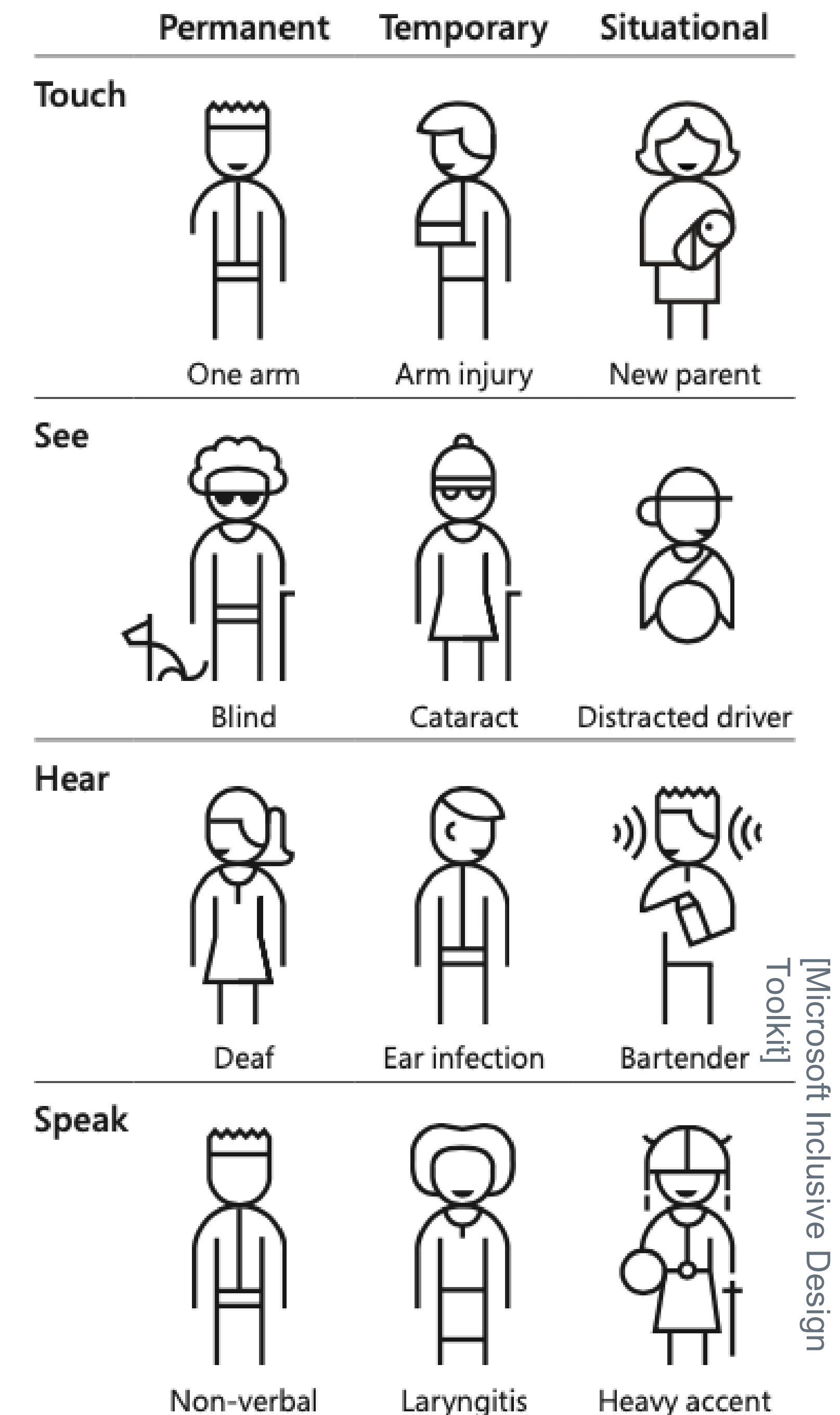
Criticized for seeing impairment as most important part of the person, and that they're helpless to do anything about it.

Focuses on **interventions for individuals**: promotes **technosolutionism**.

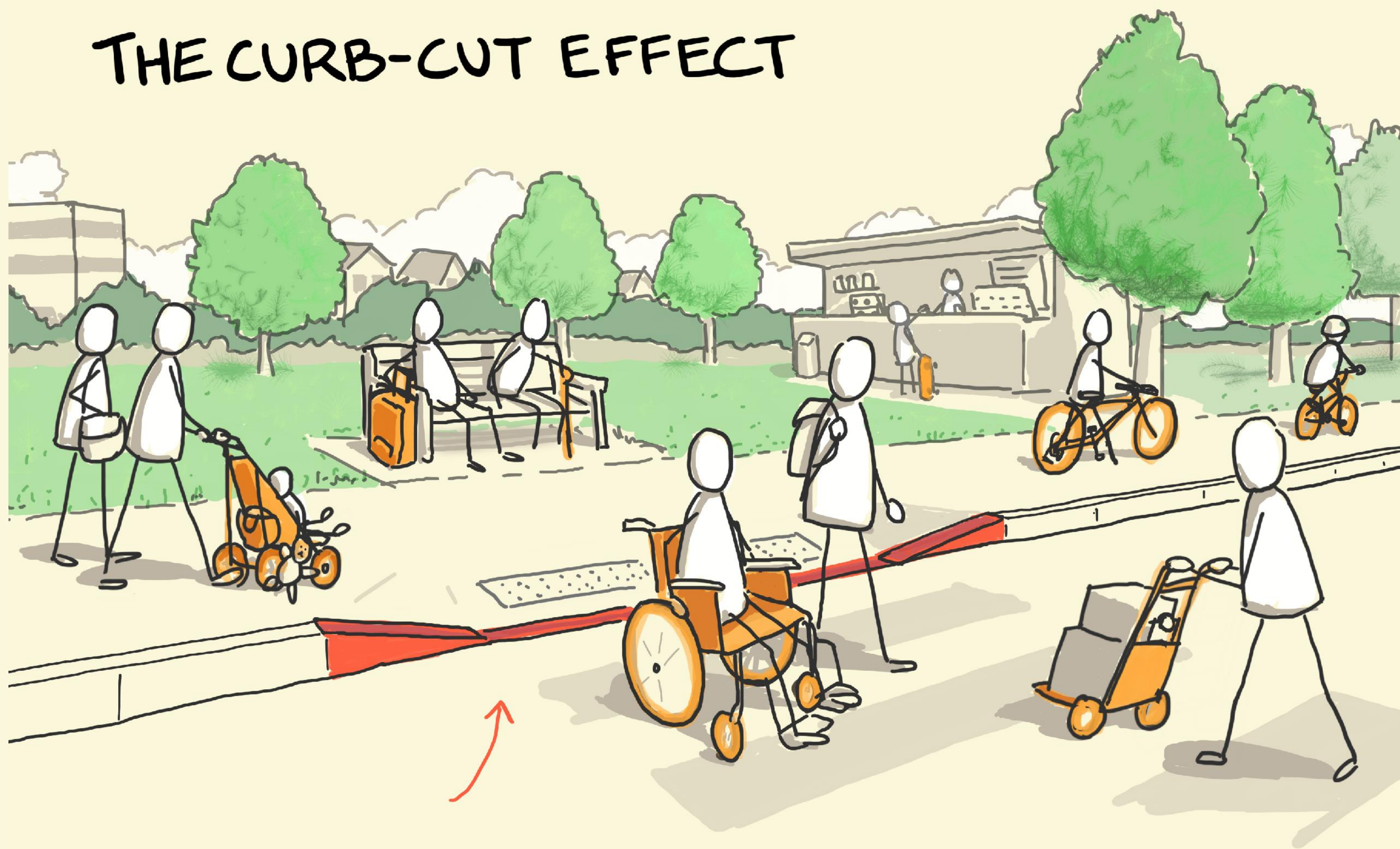
Social model of disability: people with impairments are being disabled by society's choices and attitudes. They are not having their needs met.

Recognizes that **all technology is assistive**.

Focuses on interventions on **infrastructure**.



THE CURB-CUT EFFECT



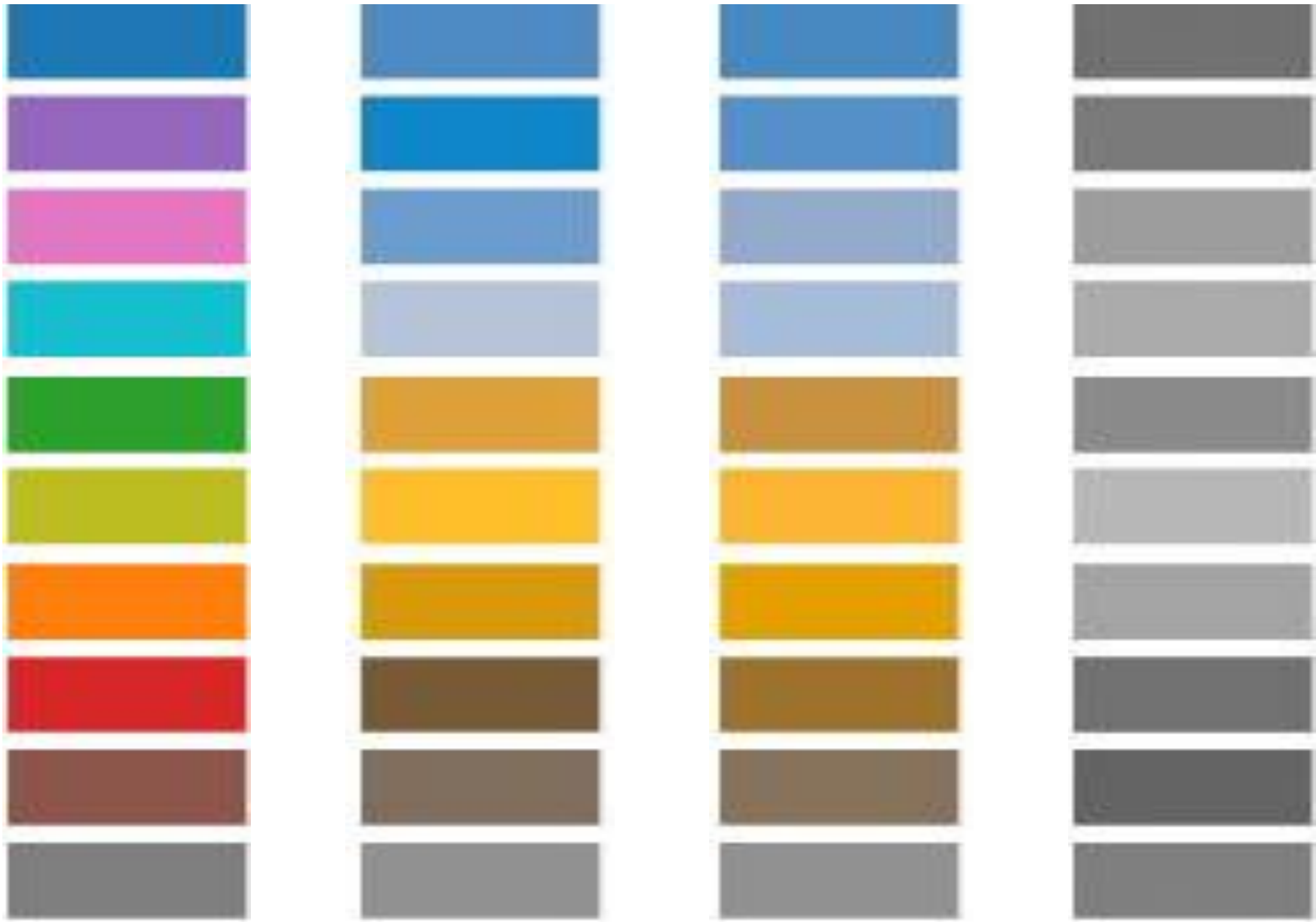
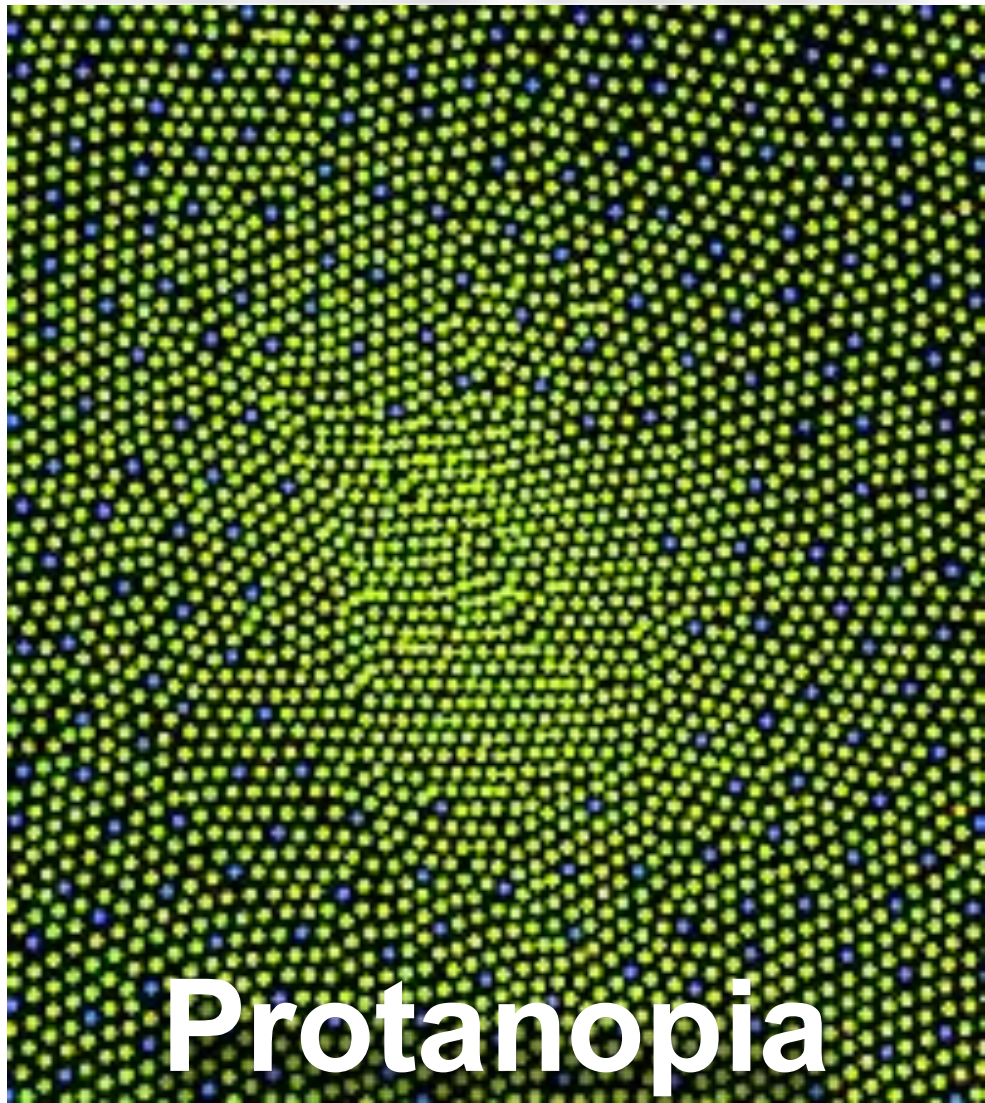
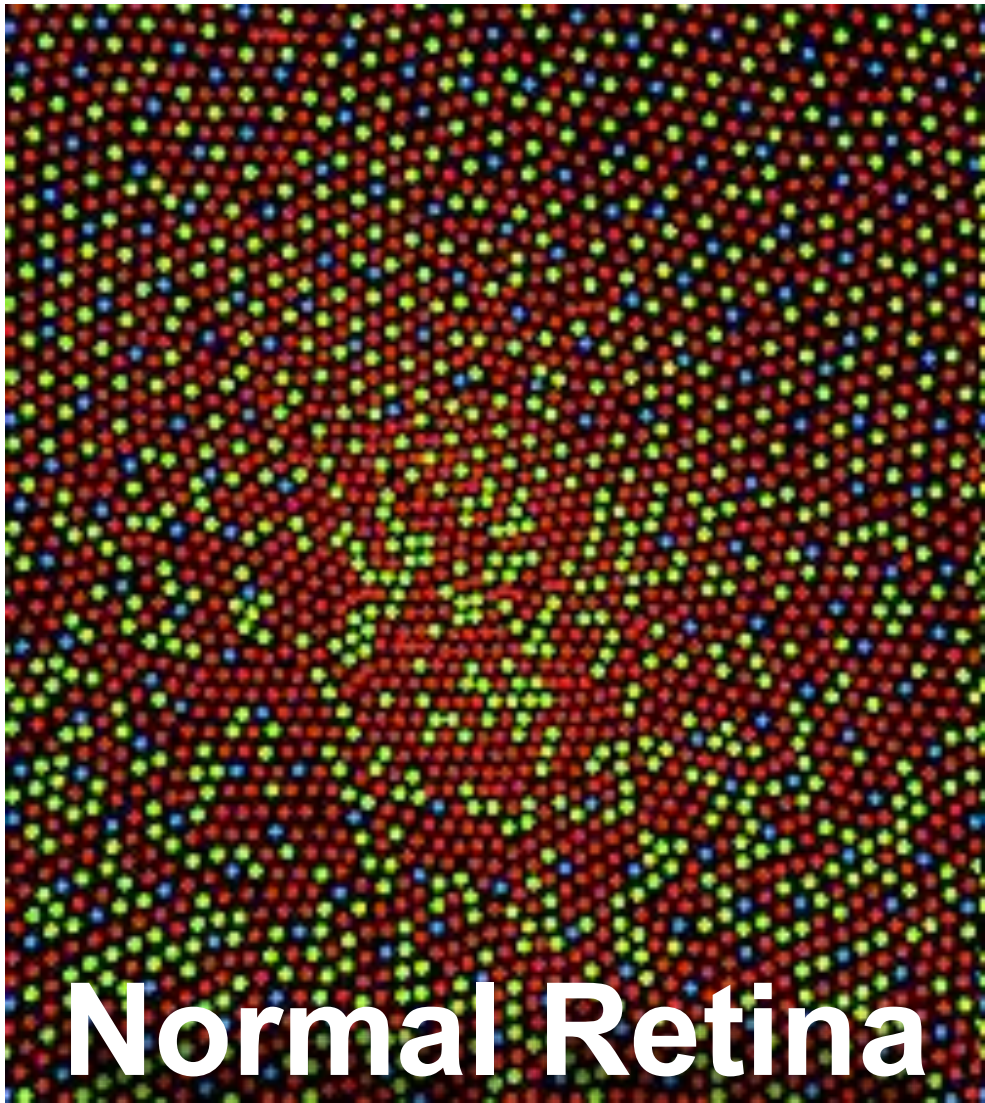
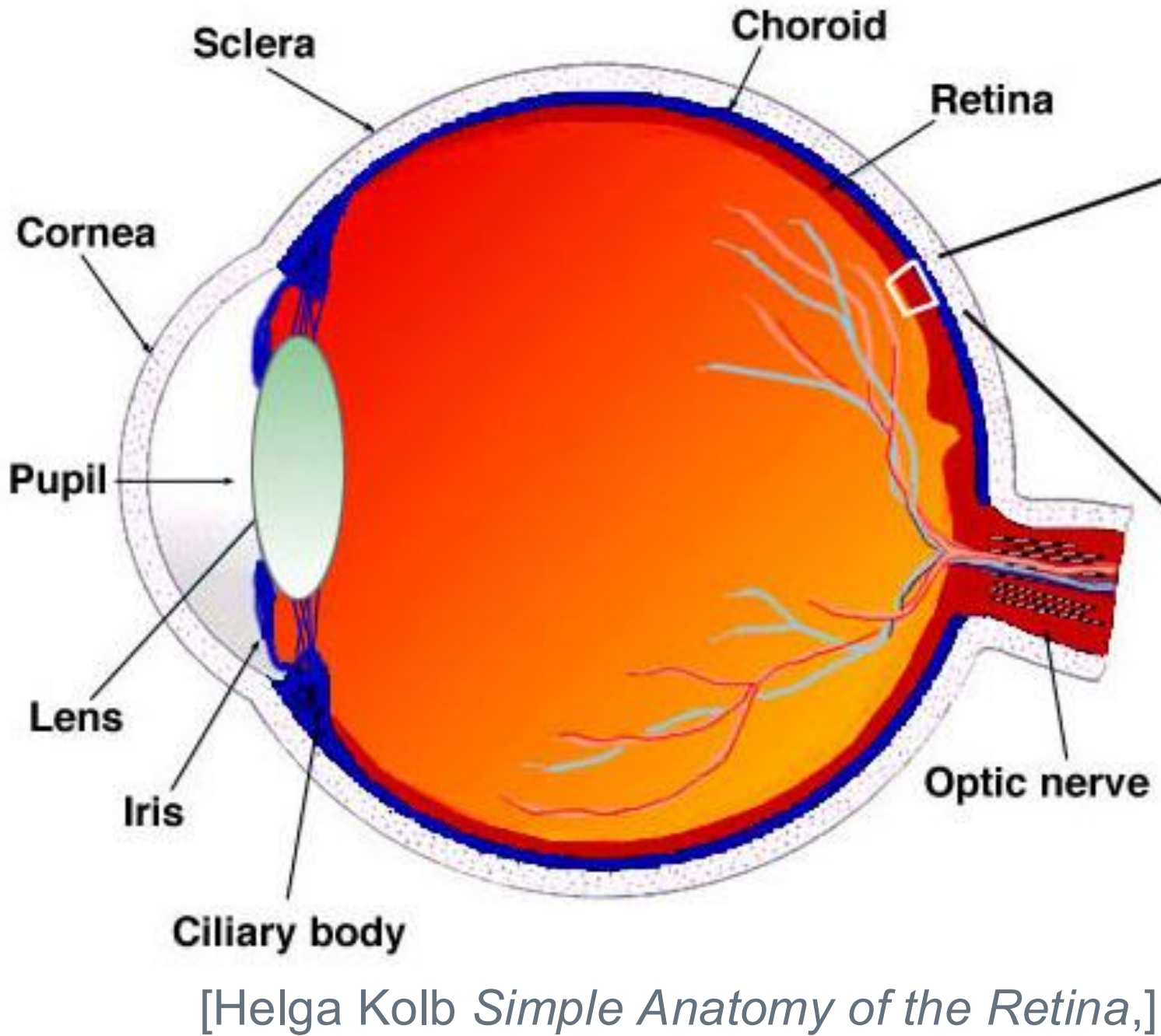
WHEN WE DESIGN
FOR DISABILITIES

... WE MAKE THINGS
BETTER FOR EVERYONE

sketchplanations

Color Blindness

Firefox and Chrome have built in simulators



Protanope

Deuteranope

Luminance

Color Schemes / D3 | Observ

observablehq.com/@d3/color-schemes

ProductLearnCommunityPricing

SearchSign inSign up

Diverging

BrBG

PRGn

PiYG

PuOr

RdBu

RdGy

RdYlBu

ElementsConsoleRecorderPerformance insightsSourcesNetworkPerformanceMemoryApplicationSecurityLighthouseComponents

topFilterAll levelsNo Issues

07:31:58.444 ▶ Fetch finished loading: POST "https://o299802.ingest.sentry.io/api/5593183/envelope/?sentry_key=ca9eac14b...bb3c08c45&sentry_version=7&sentry_client=sentry.javascript.nextjs%2F7.80.0"._app-8d4ce4ee1babf1c5.js:11

07:33:28.638 [Violation] Forced reflow while executing JavaScript took 118ms

new Rendering

or-gamut feature

tion

✓ No emulation

Blurred vision

Reduced contrast

Protanopia (no red)

Deuteranopia (no green)

Tritanopia (no blue)

Achromatopsia (no color)

No emulation

Disable AVIF image format

5

W3C Recommendation

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Later Versions of Accessibility Guidelines

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1.1.1 Non-text Content

1.2 Time-based Media

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1.2.2 Captions (Prerecorded)

1.2.3 Audio Description or Media Alternative (Prerecorded)

1.2.4 Captions (Live)

1.2.5 Audio Description (Prerecorded)

1.2.6 Sign Language (Prerecorded)

1.2.7 Extended Audio Description (Prerecorded)

1.2.8 Media Alternative (Prerecorded)

1.2.9 Audio-only (Live)

1.3 Adaptable

Web Content Accessibility Guidelines (WCAG) 2.2

W3C Recommendation 05 October 2023

▼ More details about this document

This version:
<https://www.w3.org/TR/2023/REC-WCAG22-20231005/>

Latest published version:
<https://www.w3.org/TR/WCAG22/>

Latest editor's draft:
<https://w3c.github.io/wcag/guidelines/22/>

History:
<https://www.w3.org/standards/history/WCAG22/>
[Commit history](#)

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<https://www.w3.org/WAI/WCAG22/implementation-report/>

Previous Recommendation:
<https://www.w3.org/TR/WCAG21/>

Editors:
[Alastair Campbell](#) (Nomensa)
[Chuck Adams](#) (Oracle)
[Rachael Bradley Montgomery](#) (Library of Congress)
[Michael Cooper](#) (W3C)
[Andrew Kirkpatrick](#) (Adobe)

Feedback:
[GitHub w3c/wcag](#) ([pull requests](#), [new issue](#), [open issues](#))

Errata:
[Errata exists.](#)

See also translations



<https://w3c.github.io/wcag/guidelines/22/>

W3C Recommendation

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§ 1. Perceivable

Information and user interface components must be presentable to users in ways they can perceive.

§ Guideline 1.1 Text Alternatives

Provide text alternatives for any non-text content so that it can be changed into other forms people need, such as large print, braille, speech, symbols or simpler language.

[Understanding Text Alternatives](#)

[How to Meet Text Alternatives](#)

§ Success Criterion 1.1.1 Non-text Content

(Level A)

All [non-text content](#) that is presented to the user has a [text alternative](#) that serves the equivalent purpose, except for the situations listed below.

[Understanding Non-text Content](#)

[How to Meet Non-text Content](#)

Controls, Input

If non-text content is a control or accepts user input, then it has a [name](#) that describes its purpose. (Refer to [Success Criterion 4.1.2](#) for additional requirements for controls and content that accepts user input.)

Time-Based Media

If non-text content is time-based media, then text alternatives at least provide descriptive identification of the non-text content. (Refer to [Guideline 1.2](#) for additional requirements for media.)

Test

If non-text content is a test or exercise that would be invalid if presented in [text](#), then text alternatives at least provide descriptive identification of the non-text content.

Sensory

 ©2024 WebAIM

Institute for Disability Research, Policy, and Practice
Utah State University
6807 Old Main Hill
Logan, UT 84322-6807
 435.797.7024

 Check Your Accessibility

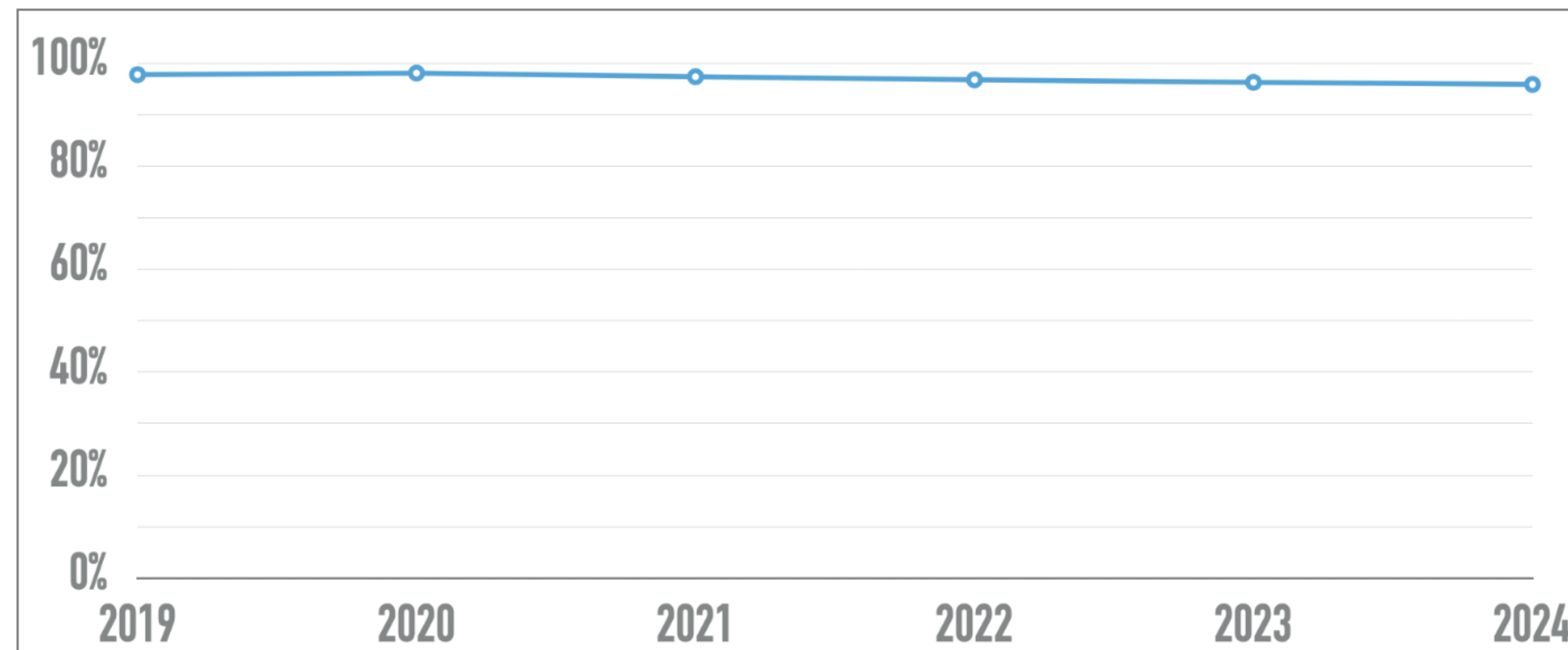
Web site address:

WAVE

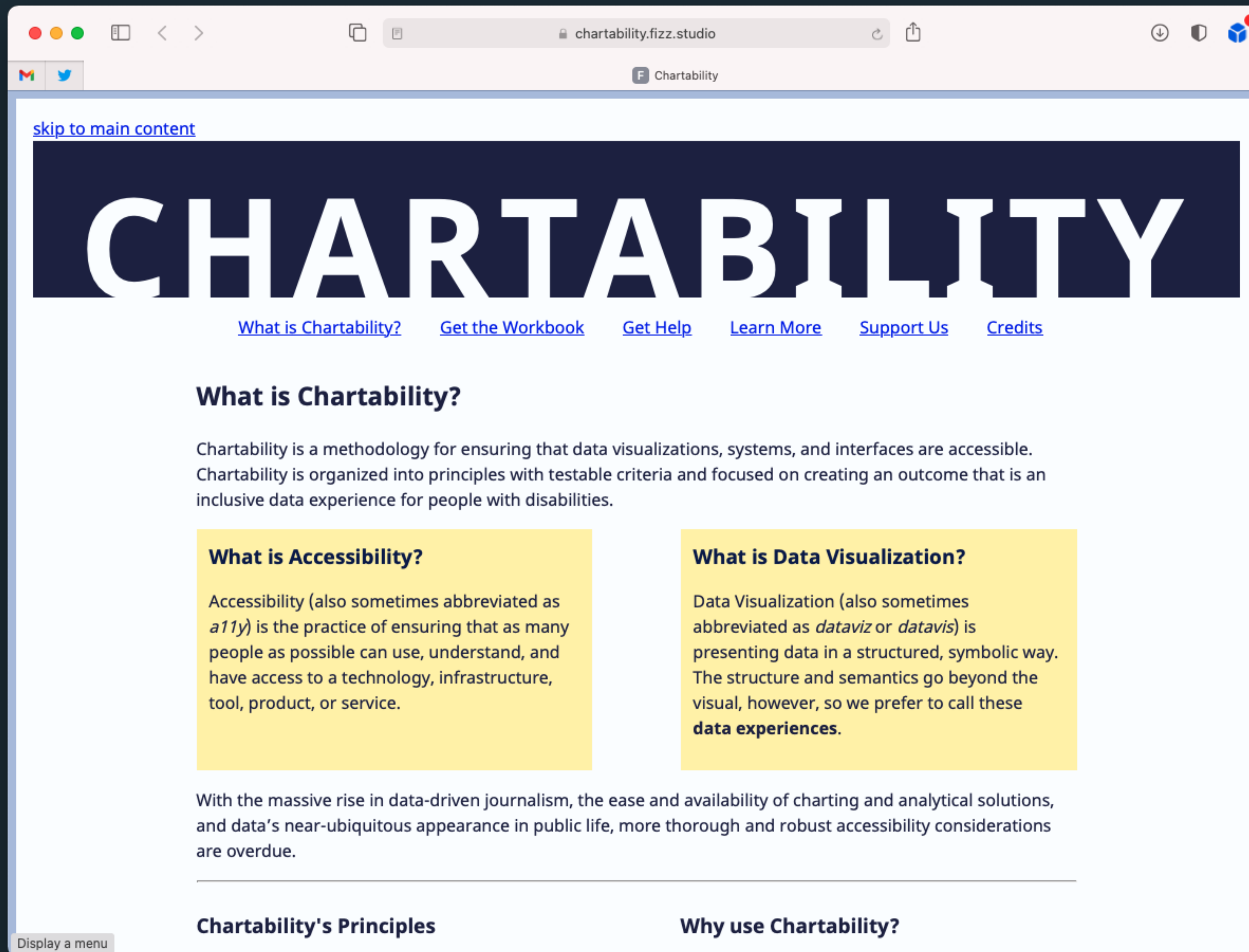
WCAG Conformance

95.9% of home pages had detected [WCAG 2](#) failures. This improved slightly from 96.3% in 2023. Over the last 5 years, the pages with detectable WCAG failures have decreased by only 1.9% from 97.8%. These are only automatically detected errors that align with WCAG conformance failures with a high level of reliability which suggests that the rate of full WCAG 2 A/AA conformance was certainly lower.

This chart shows the percentage of home pages with detected WCAG conformance failures over time:



While the rate of pages with no detectable errors was very low, 22.2% of pages had 5 or fewer detected errors and 31.2% had 10 or fewer. **Over the last several years the proportion of pages with fewer errors has increased while the number of pages with many errors has also increased—pages with fewer errors have gotten better while pages with many errors have gotten worse.**



Perceivable

Users must be able to easily identify content using their senses: sight, sound, and touch.

Operable

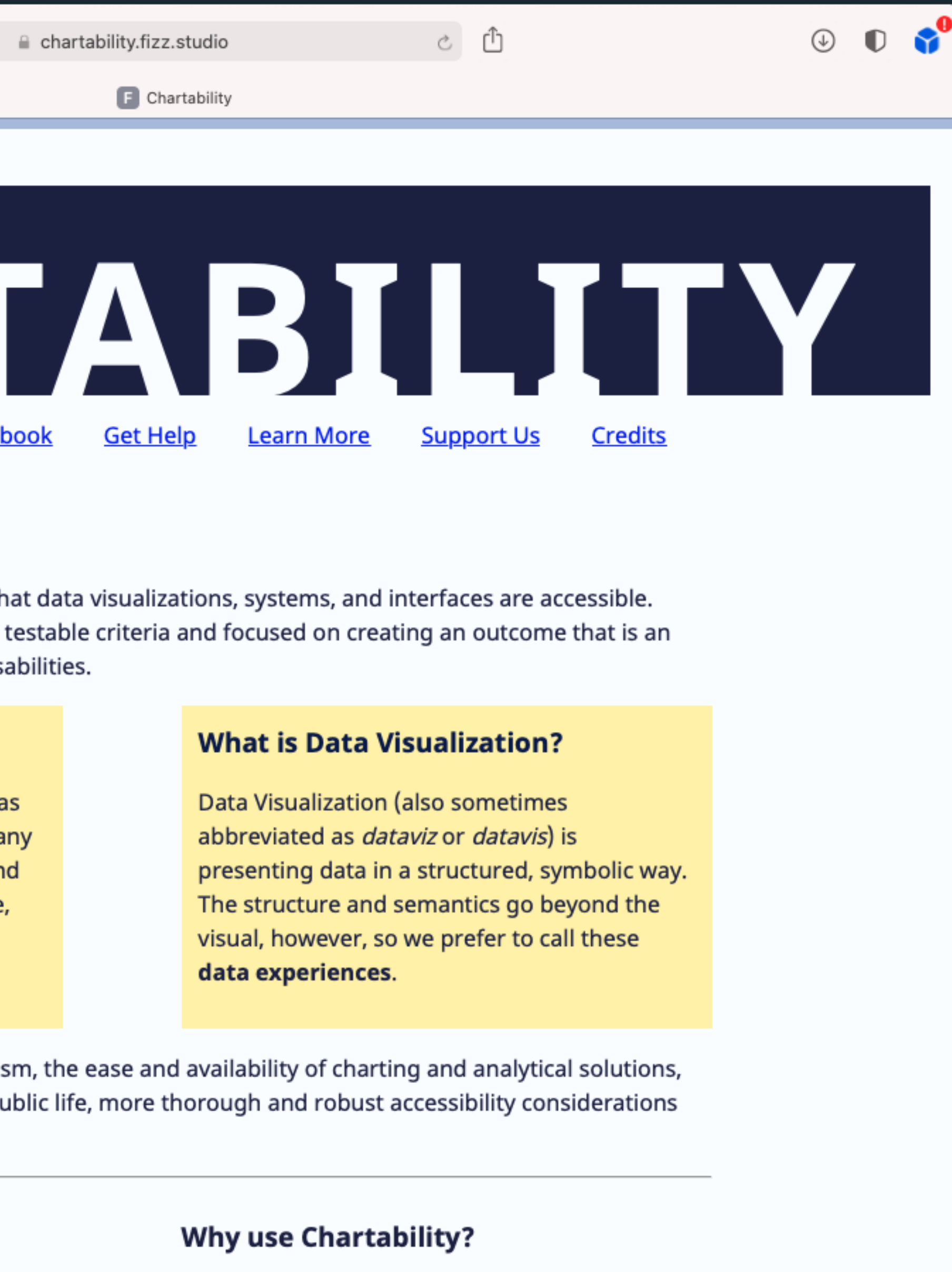
All controls must be error-tolerant, discoverable, and multi-modal (not just mouse operable, but using keyboard, etc).

Understandable

Any information or data are presented clearly, without ambiguity, and in a way that minimizes cognitive load.

Robust

The design is compliant with existing standards and works with the user's compliant, assistive technologies of choice.



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Compromising

(Understandable, yet Robust): Information flows must provide transparency, tolerance, and consideration for different ways that users with assistive technologies and disabilities will prefer to consume different information.

Assistive

(Understandable, Perceivable, labor-reducing): Interface must be laid out and information conveyed to reduce the cognitive and functional labor required for use.

Flexible

(Perceivable, Operable, yet Robust): Design must respect user settings from browsers, OS, etc. and provide presentation and operation control.

The Tests

The following section contains Chartability's 7 principles and 50 heuristics, 14 of which are considered *critical*. Those new to chartability should try testing just the *critical* heuristics first, and then move on to all of them.

Perceivable

User must be able to easily identify content using their senses: sight, sound, and touch.

Low contrast (*critical*)

▼ (select to read more)

This heuristic is synthesized from standards.

Description: Geometries and large text must have >3:1 contrast against background, Regular text must have >4.5:1.

Good example: <https://observablehq.com/@frankelavsky/high-contrast-for-data-visualization-with-examples>

Example tools or testing method: <https://webaim.org/resources/contrastchecker/>

Cited standard: <https://www.w3.org/WAI/WCAG21/Understanding/non-text-contrast.html>

► Notes (select to expand)

Content is only visual (*critical*)

► (select to read more)

Small text size (*critical*)

► (select to read more)

Visual presents seizure risk (*critical*)

<https://chartability.github.io/POUR-CAF/>

PRESIDENTIAL RESULTS

Joe Biden wins election to be the 46th US President

Pennsylvania's 20 electoral votes put native son Joe Biden above the 270 needed to become the 46th President of the United States. Born in Scranton, the former vice president and longtime Delaware senator defeated Donald Trump, the first President to lose a reelection bid since George H.W. Bush in 1992.



Let's evaluate this one with Chartability.

STATE RESULTS



Show More States

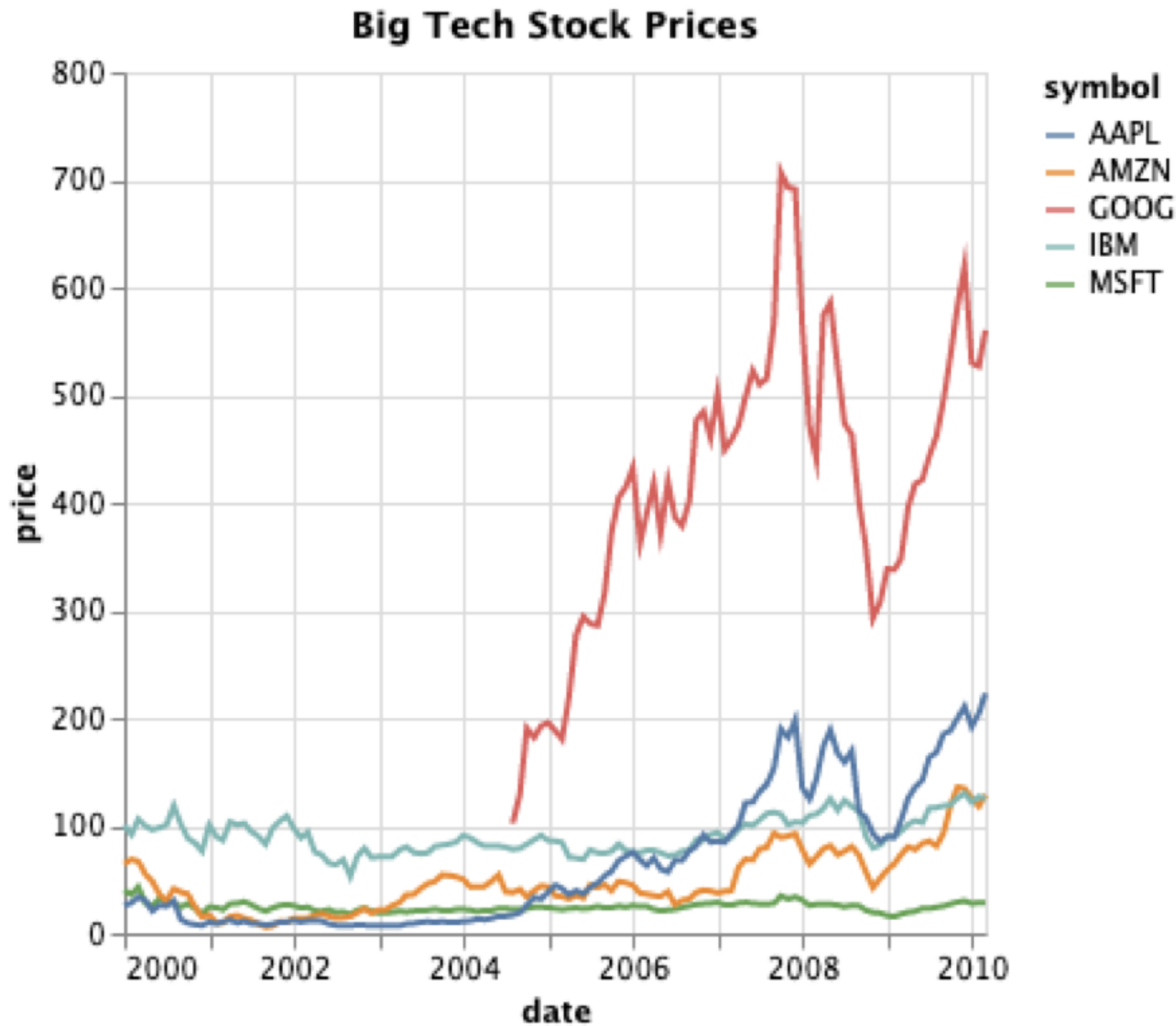
Chartability, Elan



Captions and Alt-Text

#	LEVEL KEYWORDS	SEMANTIC CONTENT	COMPUTATIONAL CONSIDERATIONS
4	contextual and domain-specific	<i>causal inferences and explanations, cultural and political relationships, current events</i>	<i>reader-dependent:</i> contextual knowledge and domain-specific insights
3	perceptual and cognitive	<i>complex trends and patterns, “natural” articulation, commonplace concepts</i>	<i>reader-agnostic:</i> reference to the rendered visualization and “common knowledge”
2	statistical and comparative	<i>descriptive statistics, extrema, outliers, correlations, point-wise comparisons</i>	<i>reader-independent:</i> access to the visualization specification and backing dataset
1	elemental and encoded	<i>chart type, encoding channels, title, axis ranges, labels, colors</i>	<i>reader-independent:</i> access to the visualization specification or rasterized image

[Alan Lundgard & Arvind Satyanarayan, in submission to VIS 2021]

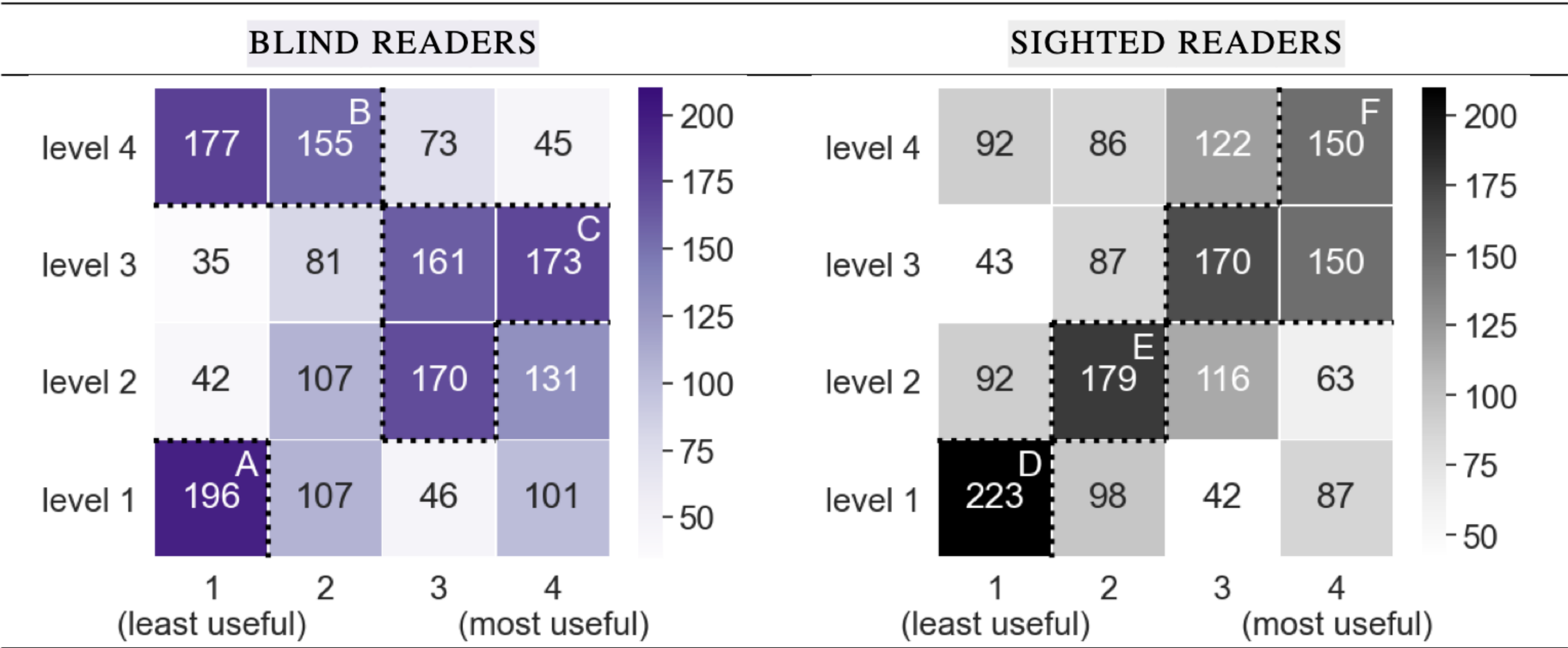


- [1] This is a multi-line chart entitled “Big Tech Stock Prices” that plots price by date. The corporations include AAPL (Apple), AMZN (Amazon), GOOG (Google), IBM (IBM), and MSFT (Microsoft). The years are plotted on the horizontal x-axis from 2000 to 2010 with an increment of 2 years. The prices are plotted on the vertical y-axis from 0 to 800 with an increment of 200. [2] GOOG has the greatest price over time. MSFT has the lowest price over time. [3] Prices of particular Big Tech corporations seem to fluctuate but nevertheless increase over time. Years 2008-2009 are exceptions as we can see an extreme drop in prices of all given corporations. [4] The big drop in prices was caused by financial crisis of 2007-2008. The crisis culminated with the bankruptcy of Lehman Brothers on September 15, 2008 and an international banking crisis. [5] At the beginning of 2008, every of this stock price went down, likely due to the financial crisis. [6] Then they have risen again and dropped again, more so than previously. [7] GOOG has the highest price over the years. MSFT has the lowest price over the years. [8] GOOG quickly became the richest one of the Big Tech corporations. [9] GOOG had experienced some kind of a crisis in 2009, because their prices drop rapidly, but then rebounded.

Captions and Alt-Text

#	LEVEL KEYWORDS	SEMANTIC CONTENT	COMPUTATIONAL CONSIDERATIONS
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[Alan Lundgard & Arvind Satyanarayan, in submission to VIS 2021]



nightingaledvs.com/writing-alt-text-for-data

TOPICS IN DATAVIZ

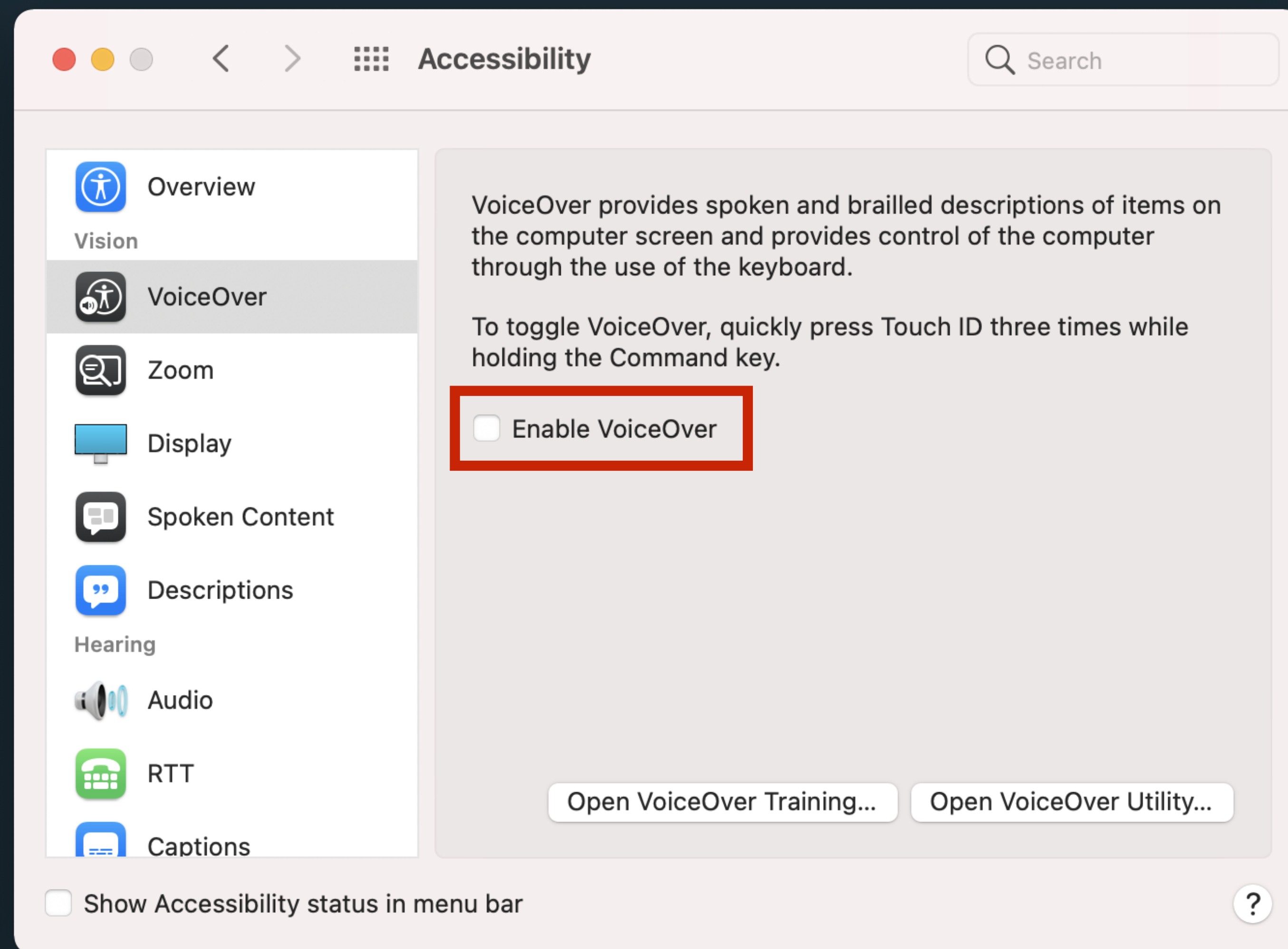
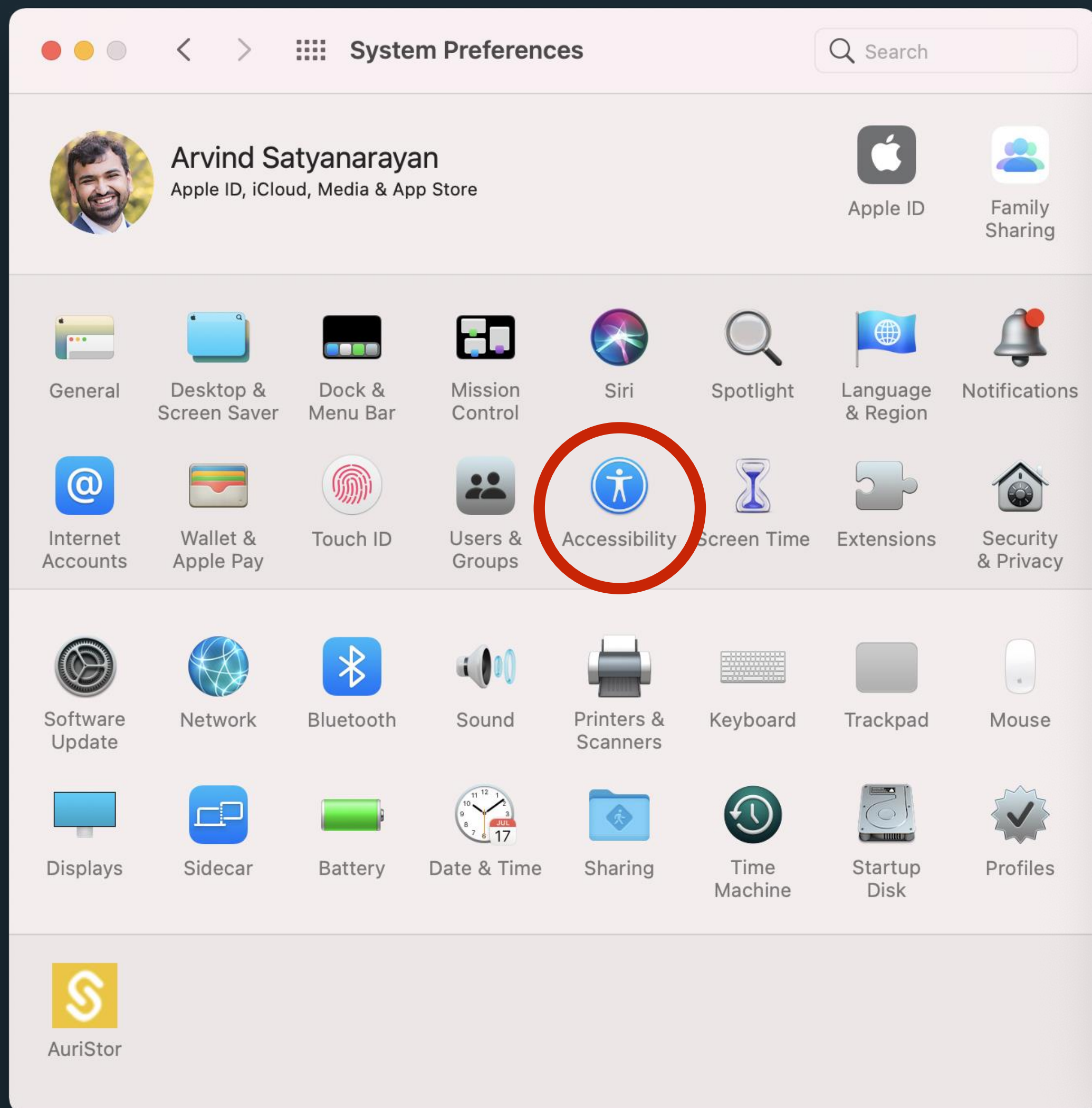
Writing Alt Text for Data Visualization

Amy Cesal • July 23, 2020

alt= “**Chart type** of **type of data**
where **reason for including chart**”

Include a **link to data source**
somewhere in the text

Alt text (sometimes called alt tags or alternative text) are written descriptions added to images that convey the meaning of the visual. Good alt text helps more people understand the content. Assistive technology like screen readers reads the alt text out loud so users can hear and understand the visual content. People who use screen readers often have a vision impairment. This includes people who are totally blind, but also people with low or impaired vision.



Accessible Rich Internet Application (ARIA)

ChromeVox extension

Case Study: Implementing Accessible Data Charts for Khan Academy by Sara Soueidan

Rich Screen Reader Experiences for Accessible Data Visualization

Jonathan Zong*
MIT CSAILCrystal Lee*
MIT CSAILAlan Lundgard*
MIT CSAILJiWoong Jang
Carnegie Mellon UniversityDaniel Hajas
Global Disability Innovation HubArvind Satyanarayan
MIT CSAIL

Computer Graphics Forum (Proc. EuroVis), 2022

 Best Paper Honorable Mention

ABSTRACT—Current web accessibility guidelines ask visualization designers to support screen readers via basic non-visual alternatives like textual descriptions and access to raw data tables. But charts do more than summarize data or reproduce tables; they afford interactive data exploration at varying levels of granularity—from fine-grained datum-by-datum reading to skimming and surfacing high-level trends. In response to the lack of comparable non-visual affordances, we present a set of rich screen reader experiences for accessible data visualization and exploration. Through an iterative co-design process, we identify three key design dimensions for expressive screen reader accessibility: *structure*, or how chart entities should be organized for a screen reader to traverse; *navigation*, or the structural, spatial, and targeted operations a user might perform to step through the structure; and, *description*, or the semantic content, composition, and veracity of the screen reader's narration. We operationalize these dimensions to prototype screen-reader-

range of chart types and combinations of our design dimensions. In a mixed-methods study with 13 blind and visually impaired participants, our designs help users conceptualize data spatially, selectively explore data at different levels of granularity, and experience control and agency over their data. The code of this paper is available at: <http://vis.csail.mit.edu/pubs/rich-screen-reader-vis-experiences>.

× Rich Screen Reader Experiences for Accessible Data Visualization | MIT Visualization Group, web content

1 INTRODUCTION

Despite decades of visualization research and recent legal requirements to make web-based content accessible [59, 30], web-based visualizations remain largely inaccessible to people with visual disabilities. Charts on mainstream publications are often completely invisible to screen readers (an assistive technology that transforms text and visual media into speech) or are rendered as incomprehensible strings of “*graphic graphic*” [54, 49]. Current accessibility guidelines ask visualization designers to provide textual descriptions of their graphics via alt text (short for alternative text) and link to underlying data tables [26, 60]. However, these recommendations do not provide modes of information-seeking comparable to what sighted readers enjoy with interactive visualizations. For instance, well-written alt text can provide a high-level takeaway of what the visualization shows, but it does not allow readers to drill down into the data to explore specific sections. While tables provide readers with the ability to hone in on specific data points, reading data line-by-line quickly becomes tedious and makes it difficult to identify overall trends.

Developing rich non-visual screen reader experiences for data visualizations poses several unique challenges. Although visuomotor interactions (like hovering, pointing, clicking, and dragging) have been core to visualization research [22], screen readers redefine what interaction is for visualization. Rather than primarily *manipulating* aspects of the visualization or its backing data pipeline [65, 29, 22], screen readers make *reading* a visualization an interactive operation as well—users must intentionally perform actions with their input

Materials

 Paper PDF Supplementary MaterialBibTeX

```
@article{2022-rich-screen-re
  title = {{Rich Screen Read
  author = {Jonathan Zong AN
  journal = {Computer Graphi
  year = {2022},
  url = {http://vis.csail.mi
}
```

Videos

Video Demo



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- 5 Evaluation
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 - 5.2 Quantitative Results
 - 5.3 Qualitative Results
- 6 Discussion and Future Work
 - 6.1 Enabling Richer Screen Reader Experiences Today



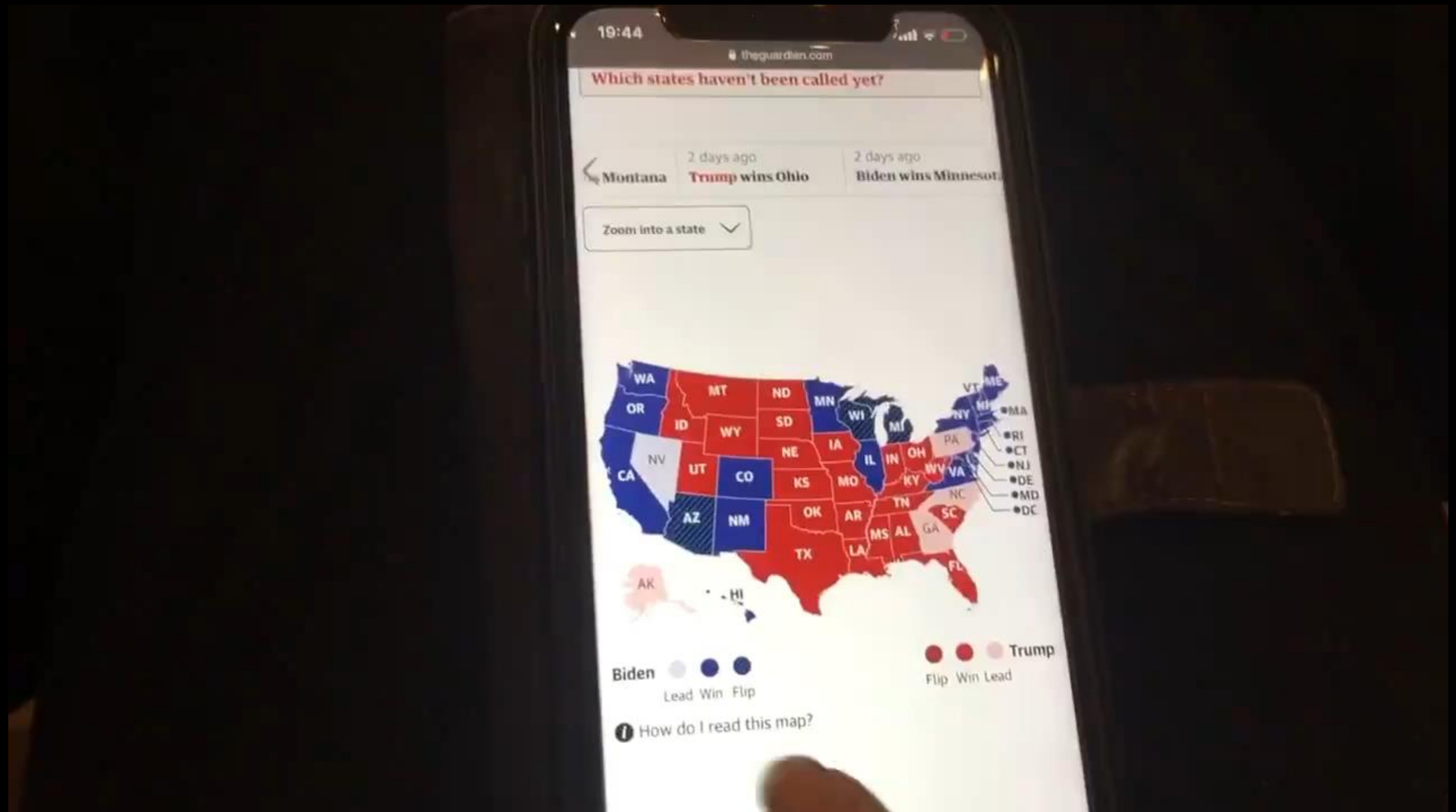




CHART ROOT

SUMMARY

“A scatterplot with X on the x-axis and Y on the y-axis; Categories ○ and × ...”

ENCODINGS

X-AXIS

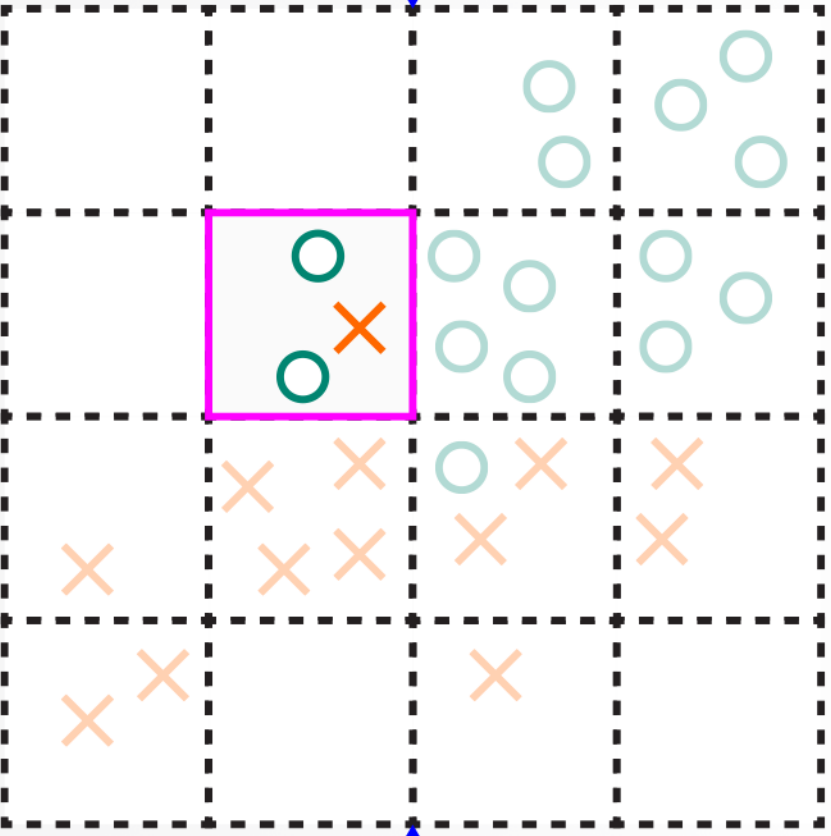
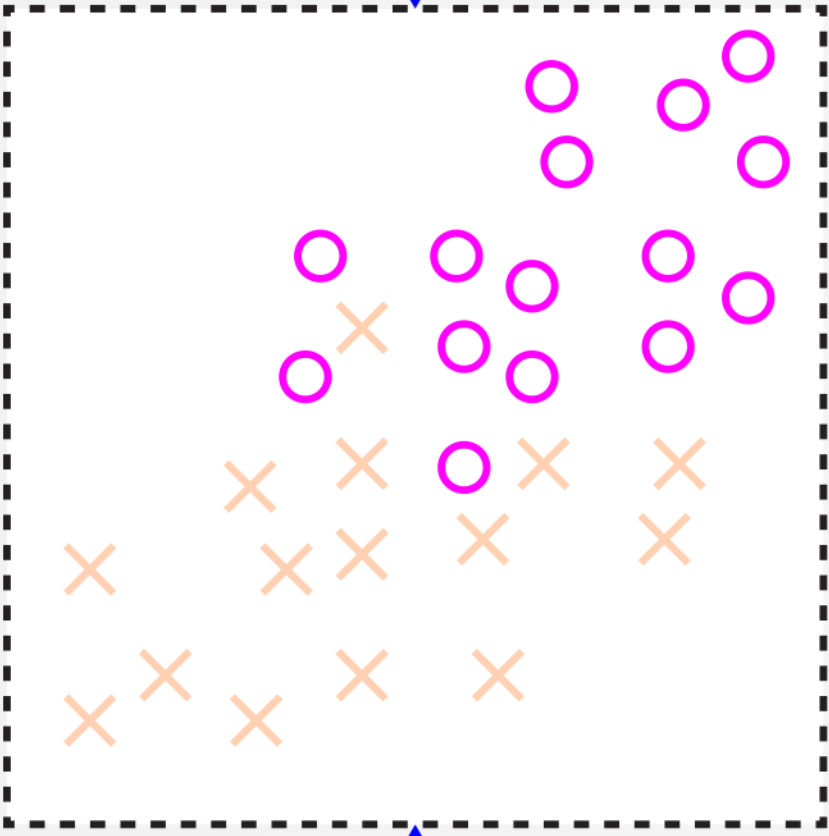
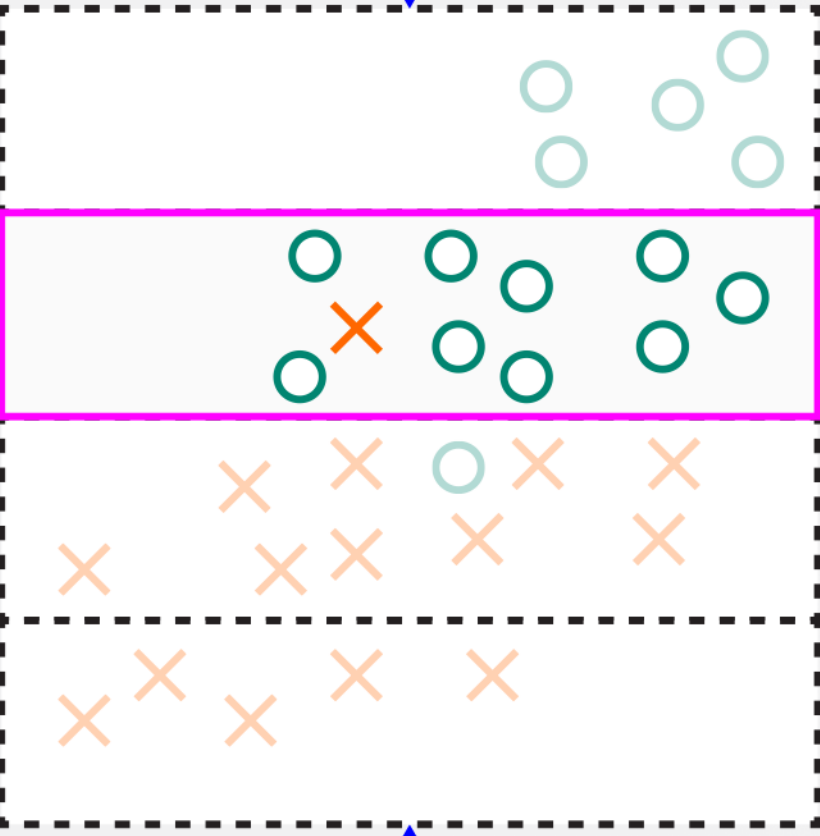
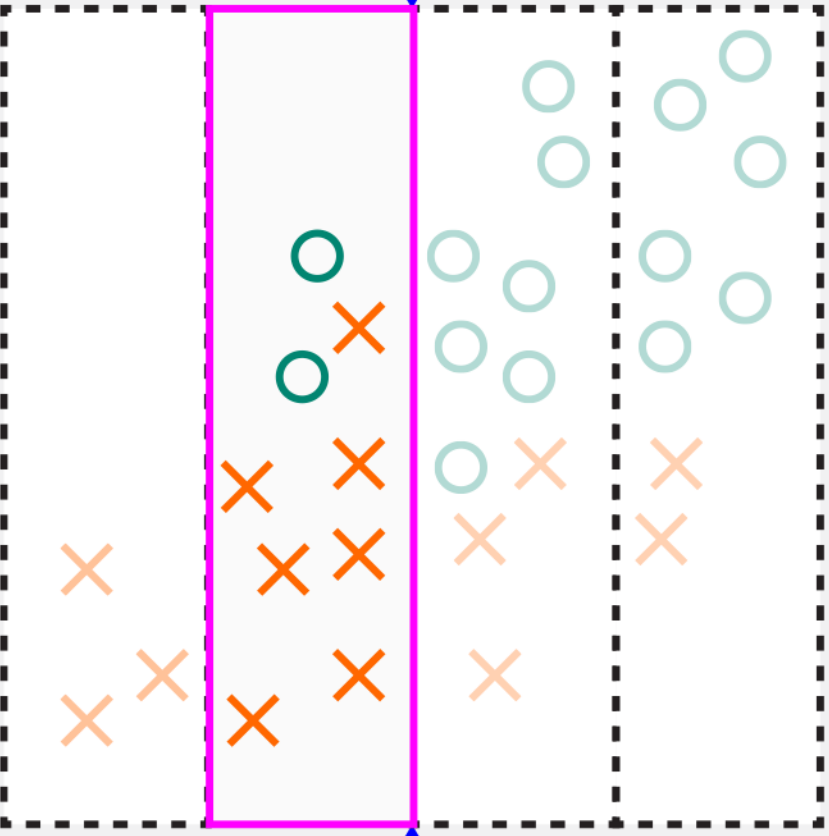
Y-AXIS

LEGEND

X-Y GRID

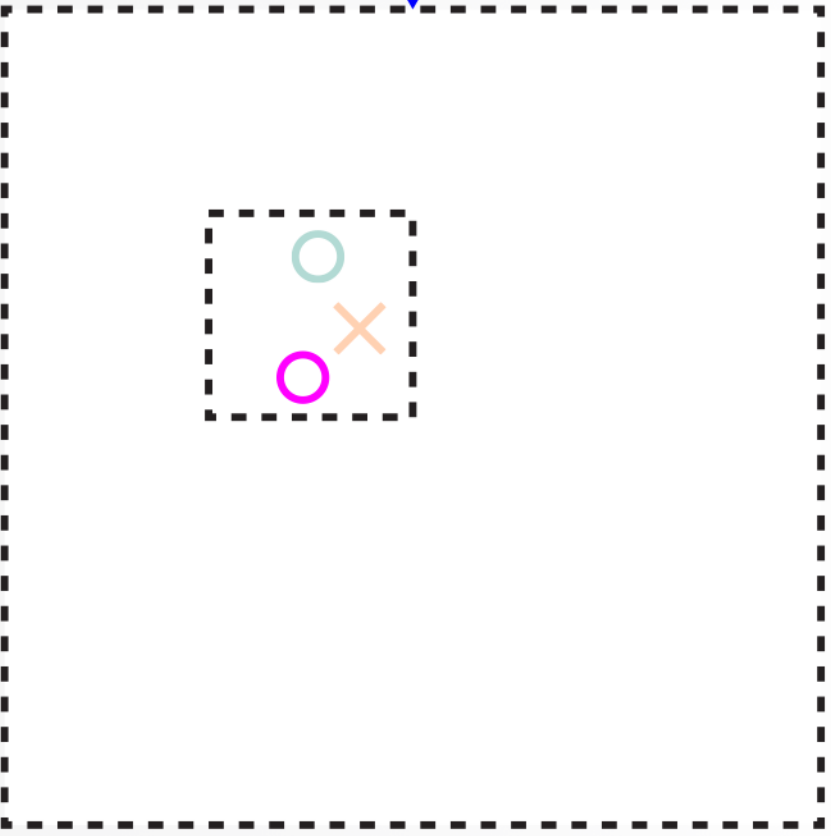
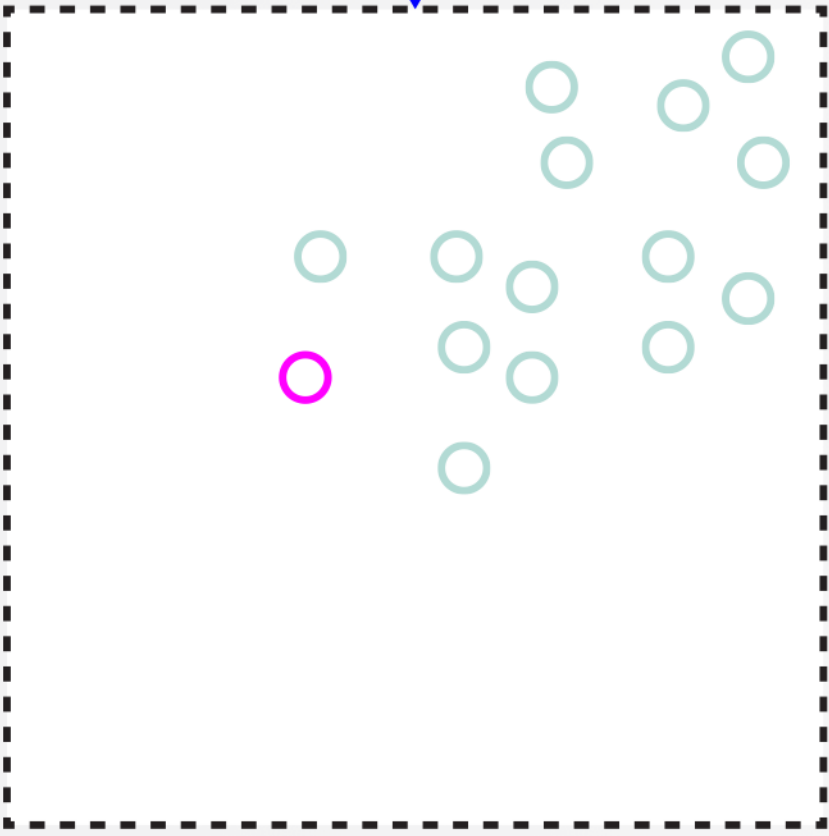
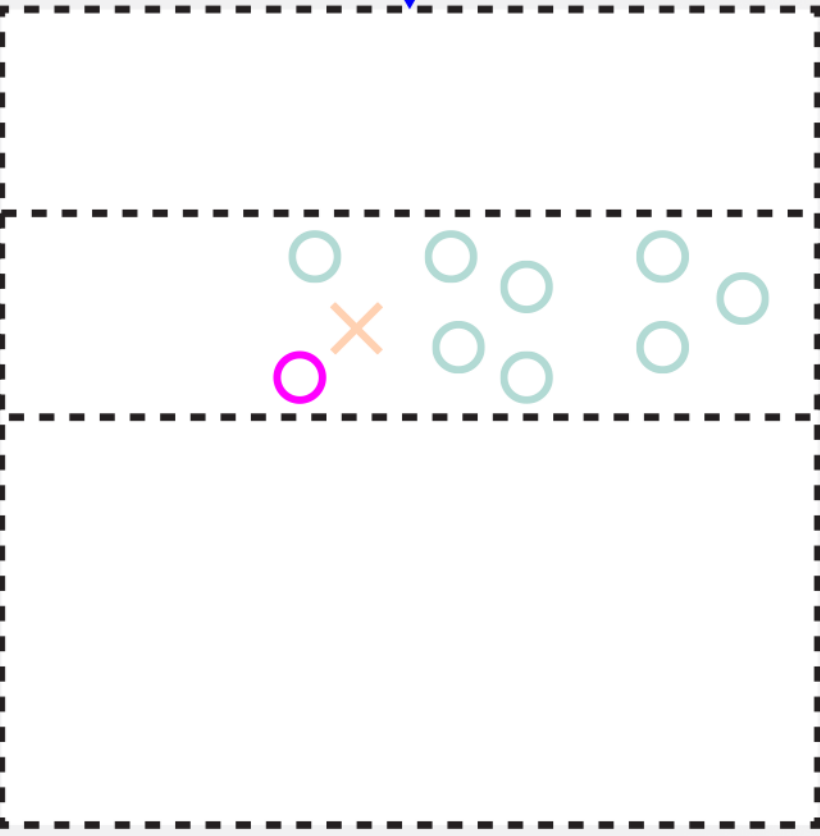
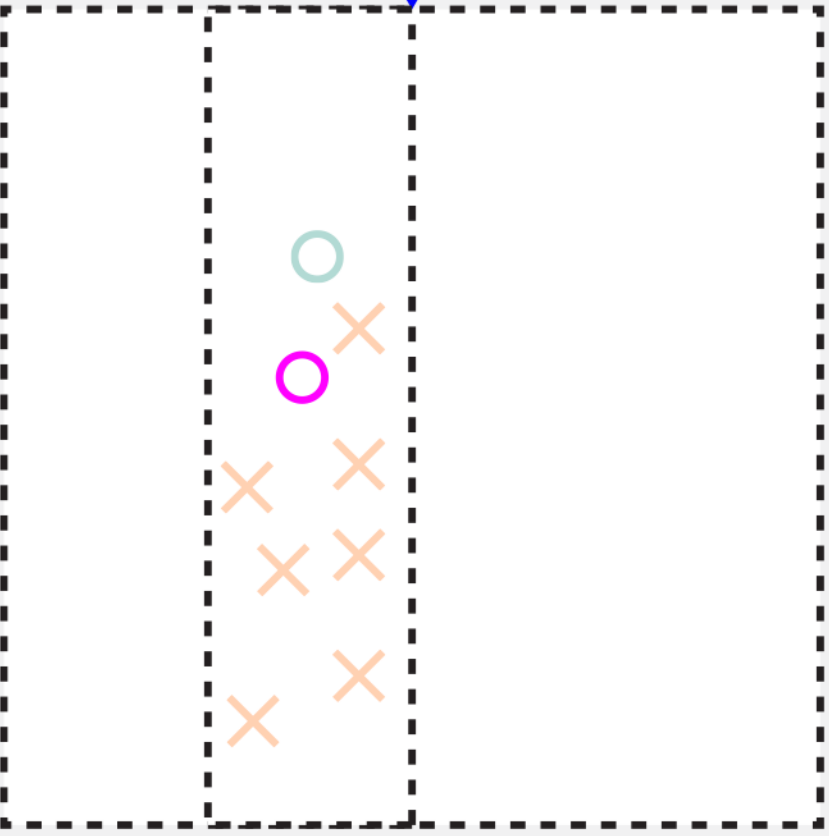
INTERVALS / CATEGORIES

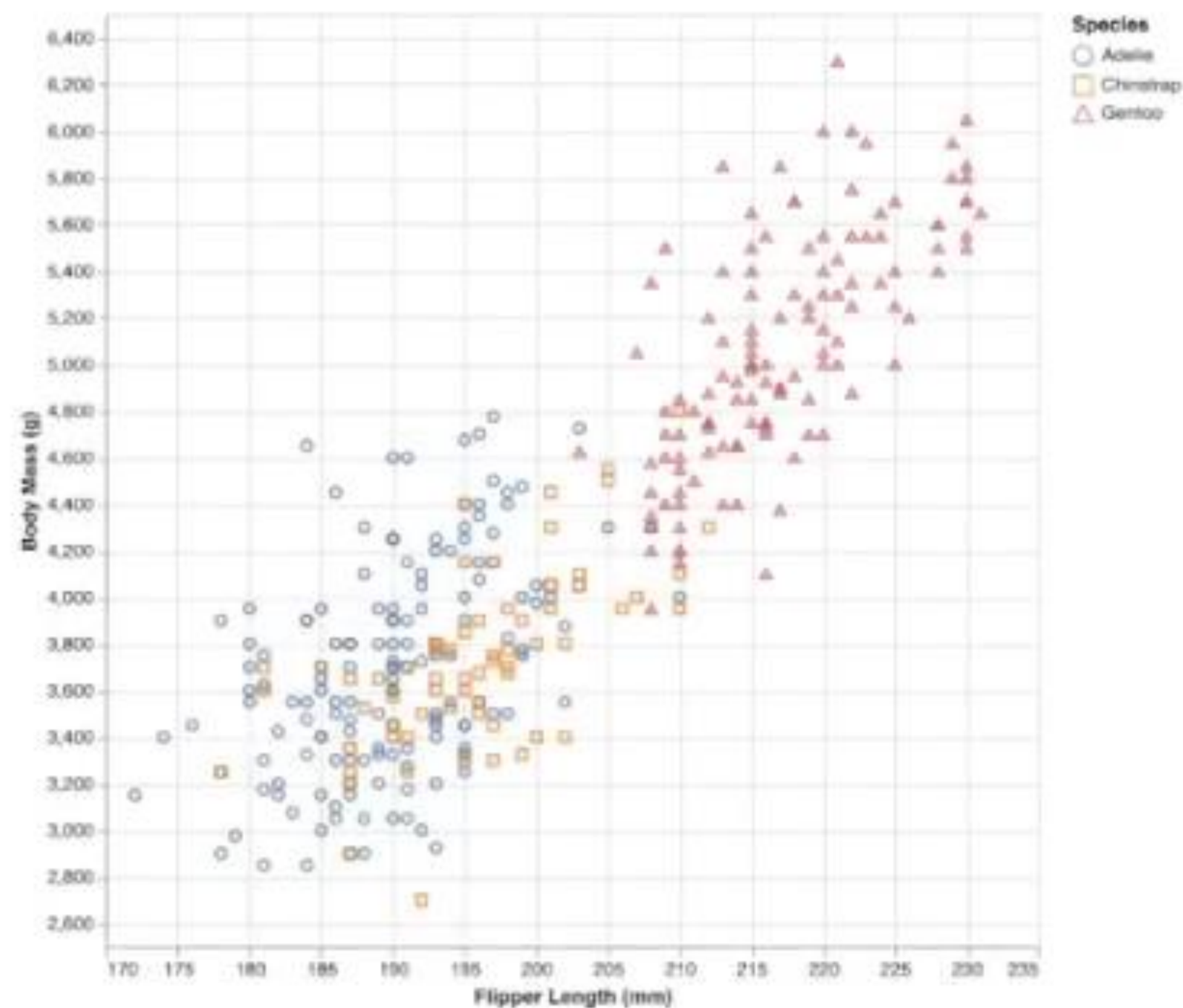
to select interval / category



DATA POINTS

to select data point





A scatterplot showing body mass and flipper lengths of penguins.



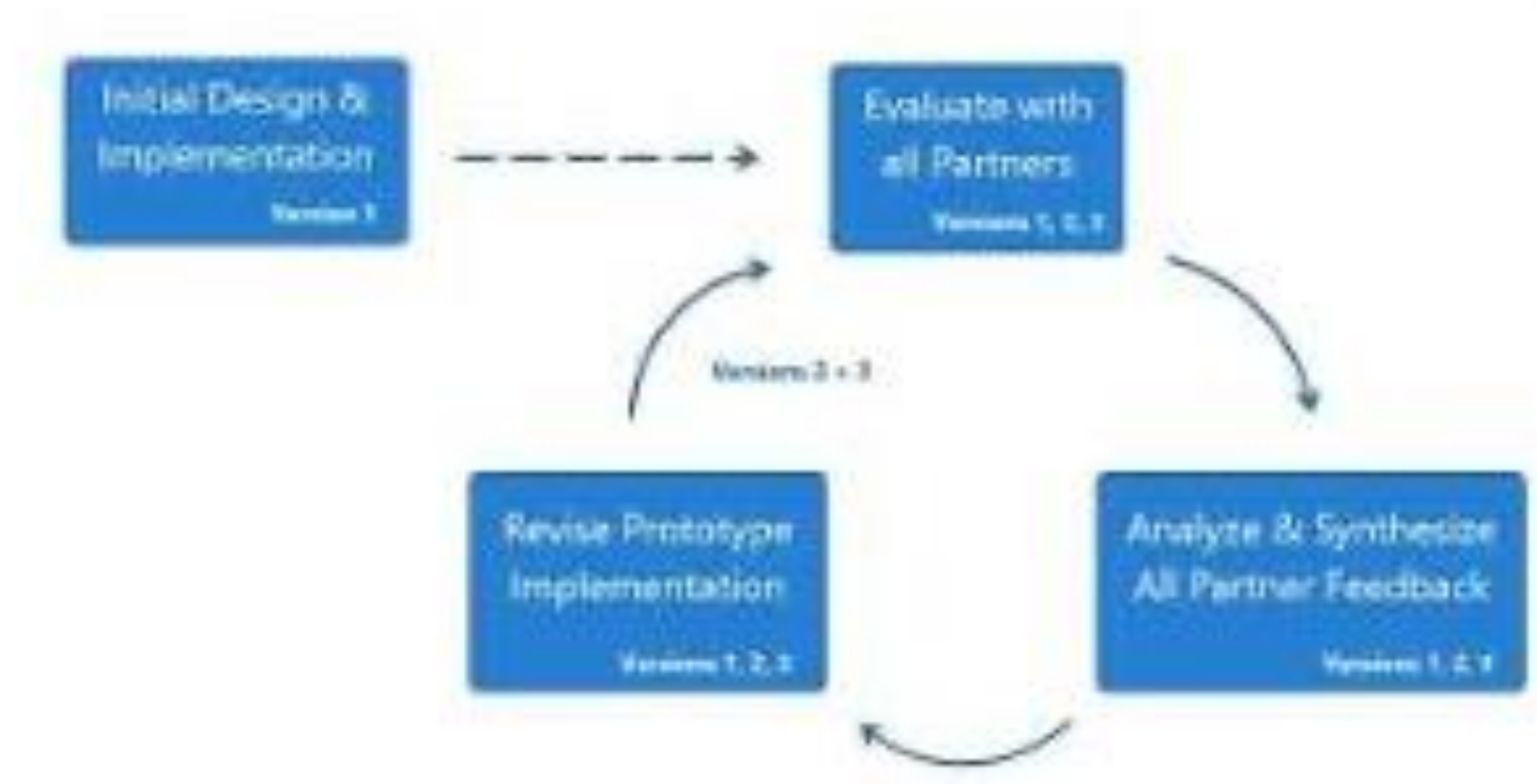
× A scatterplot showing body mass and flipper lengths of penguins.







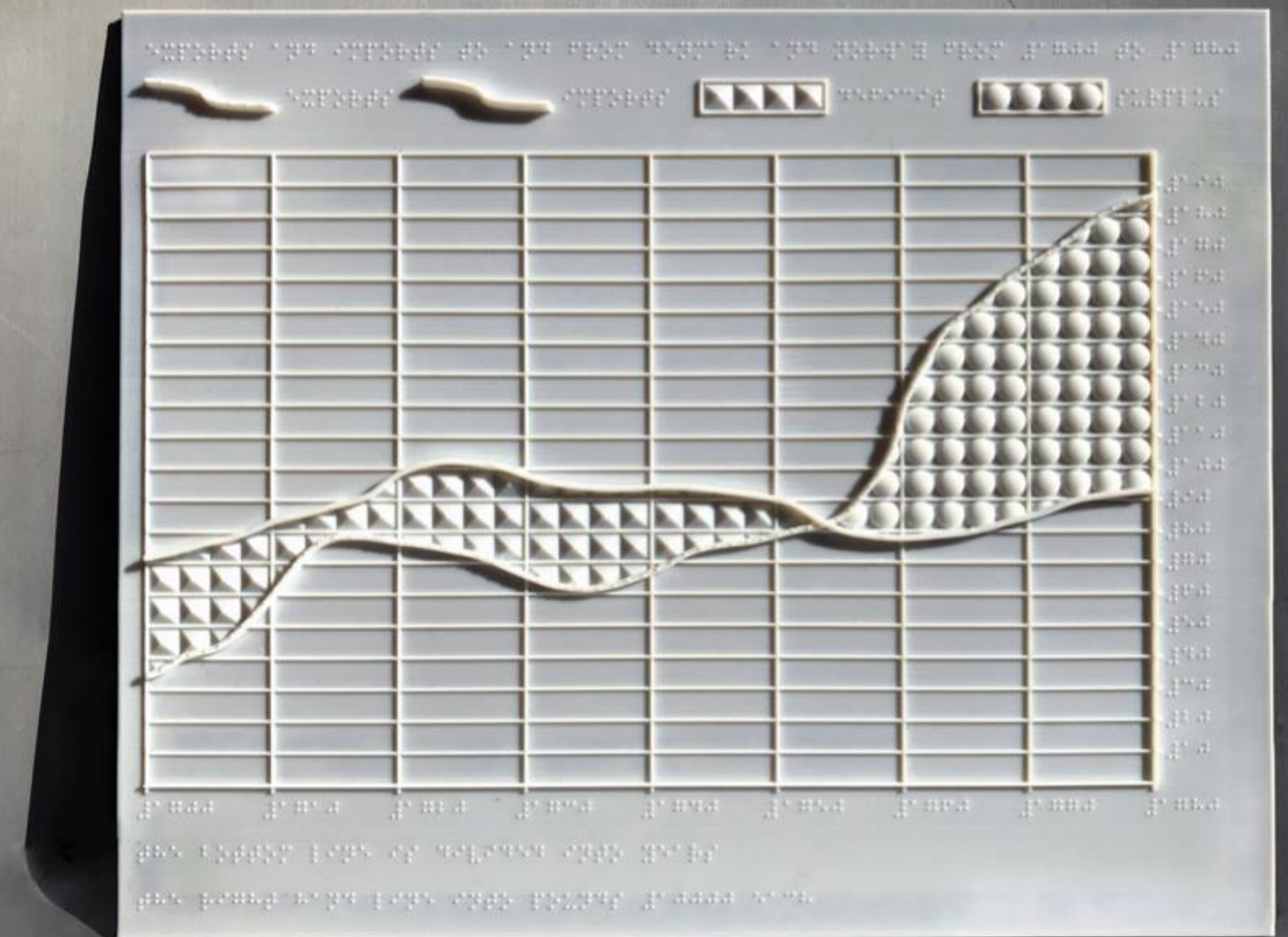
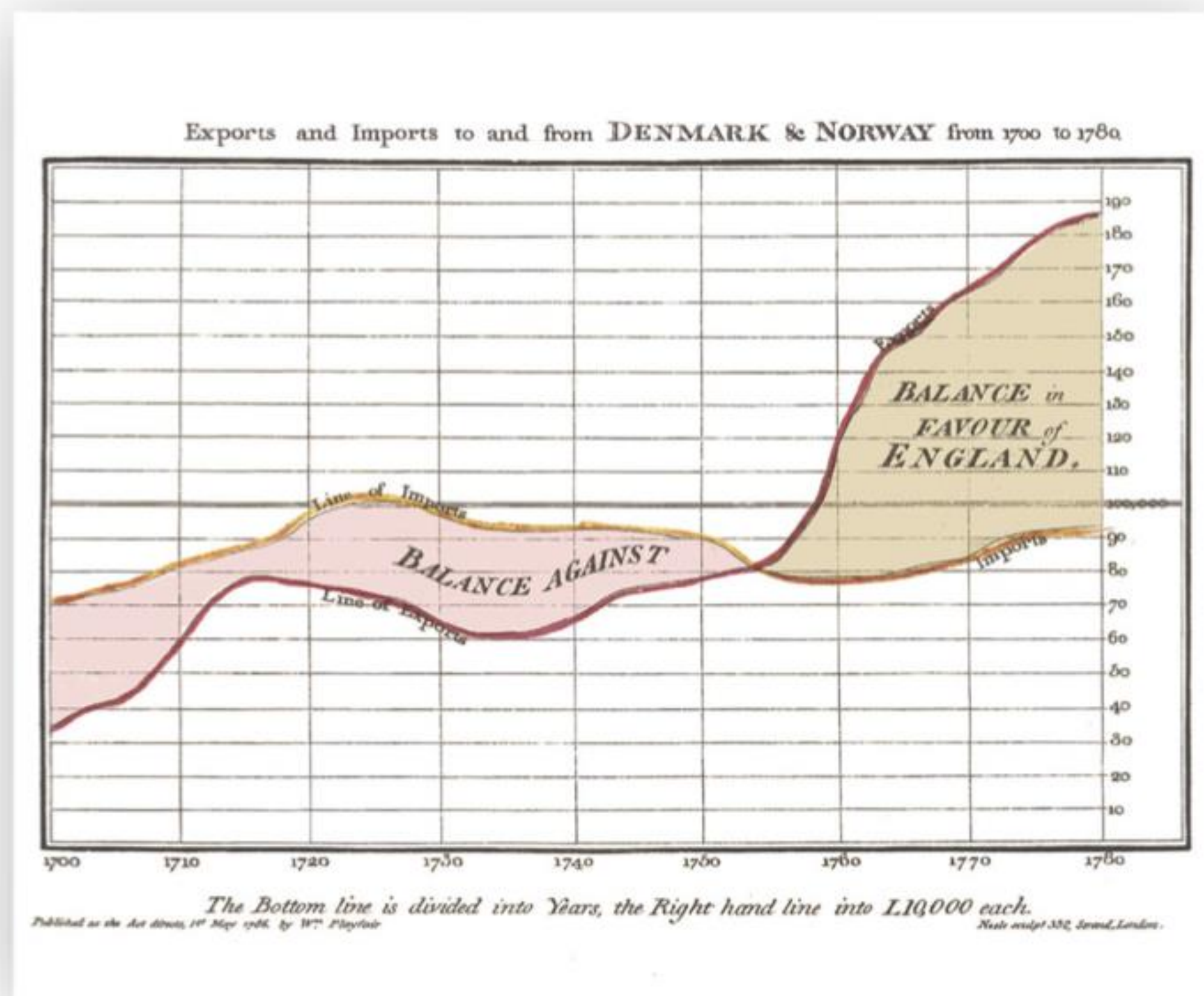
(Iterative) Participatory Design



VoxLens: Making Online Data Visualizations Accessible with an Interactive JavaScript Plug-In



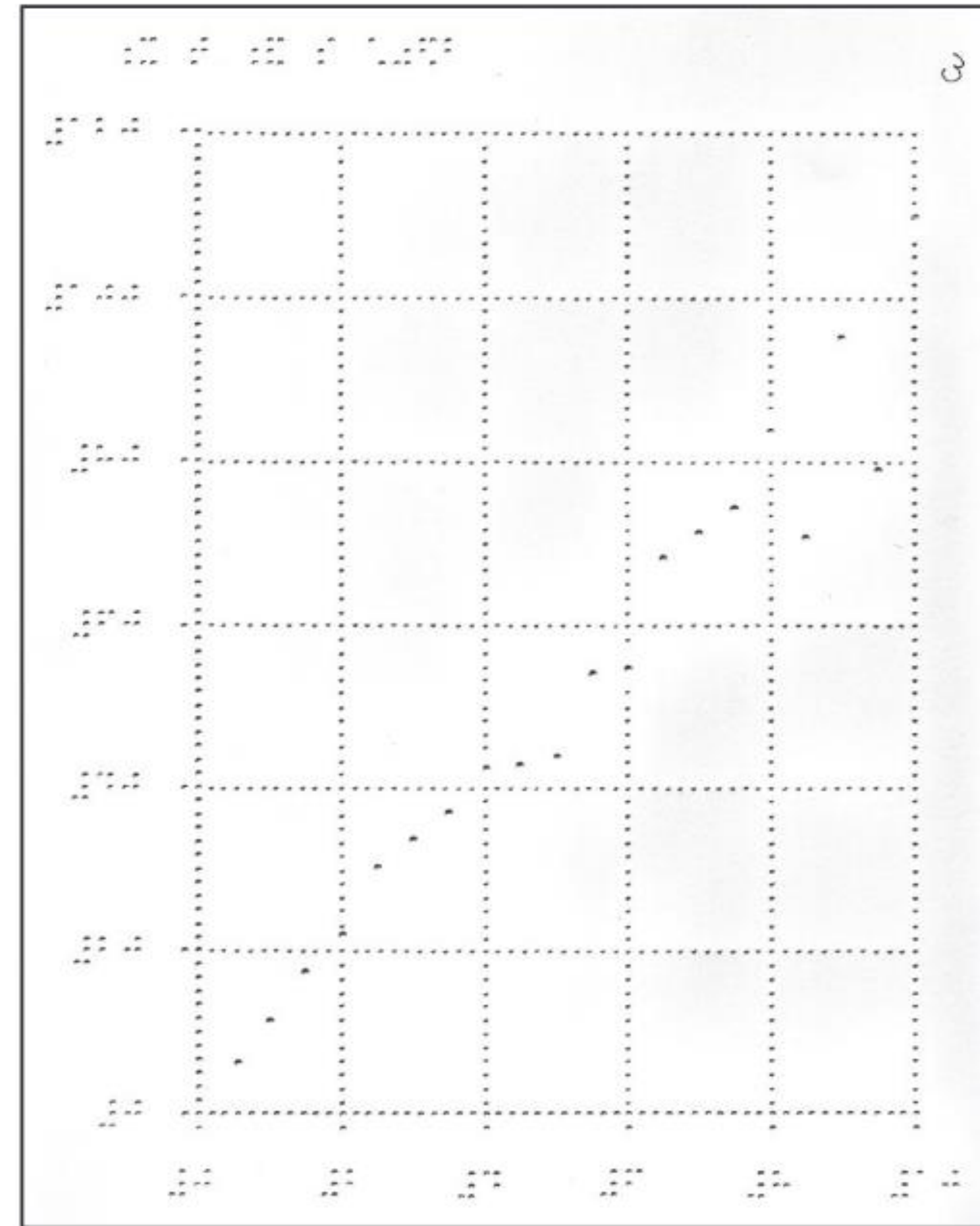
Ather Sharif, Olivia H. Wang, Alida T. Muongchan, Katharina Reinecke, and Jacob Wobbrock



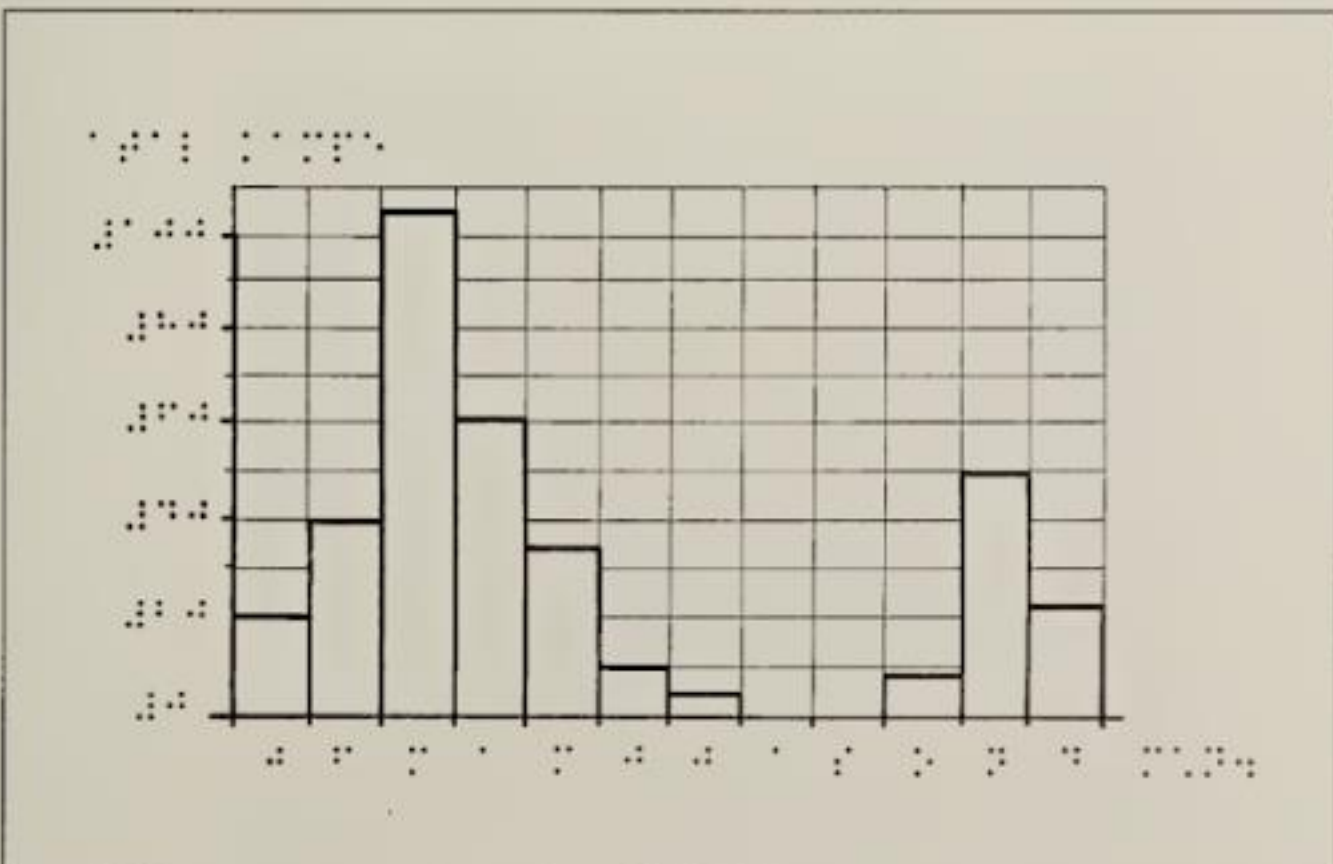
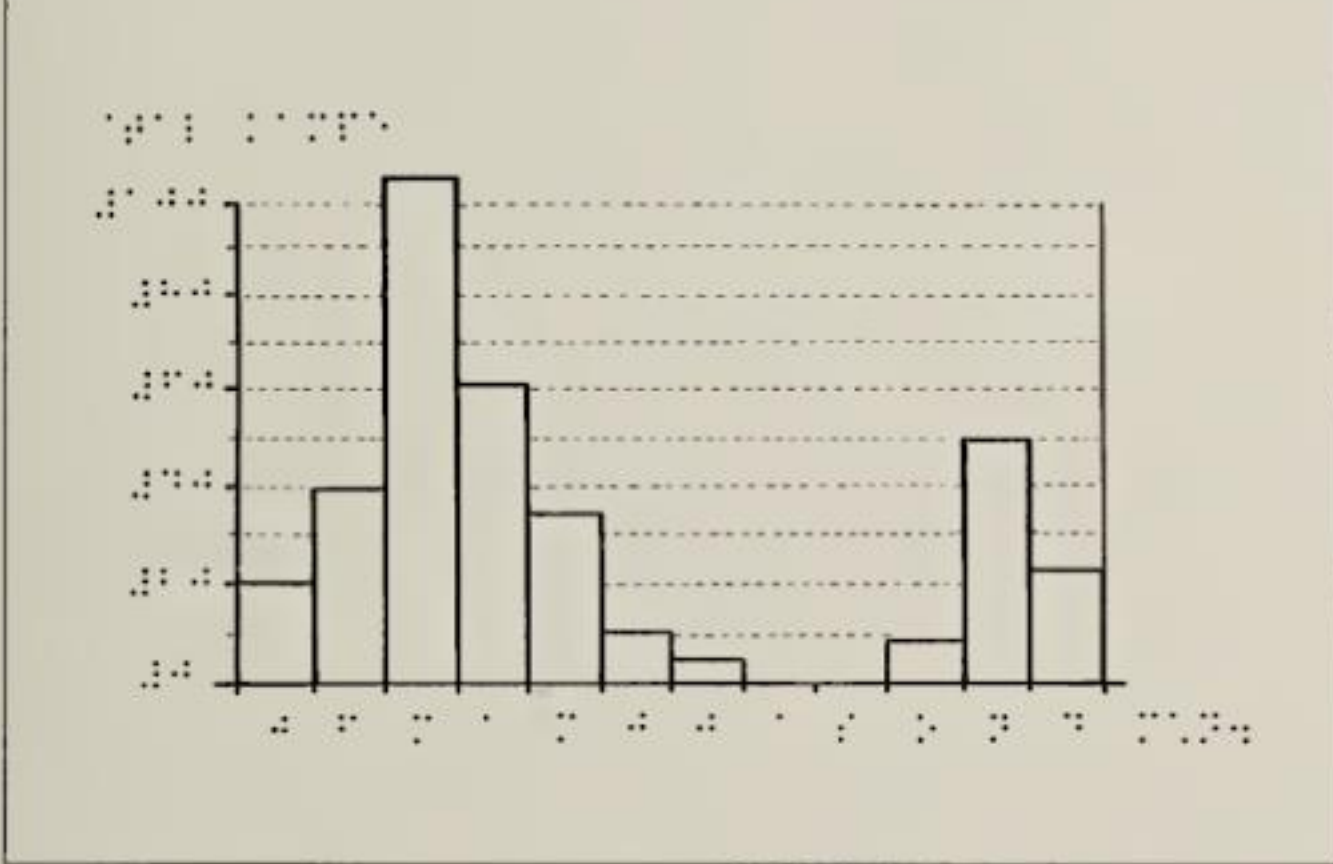
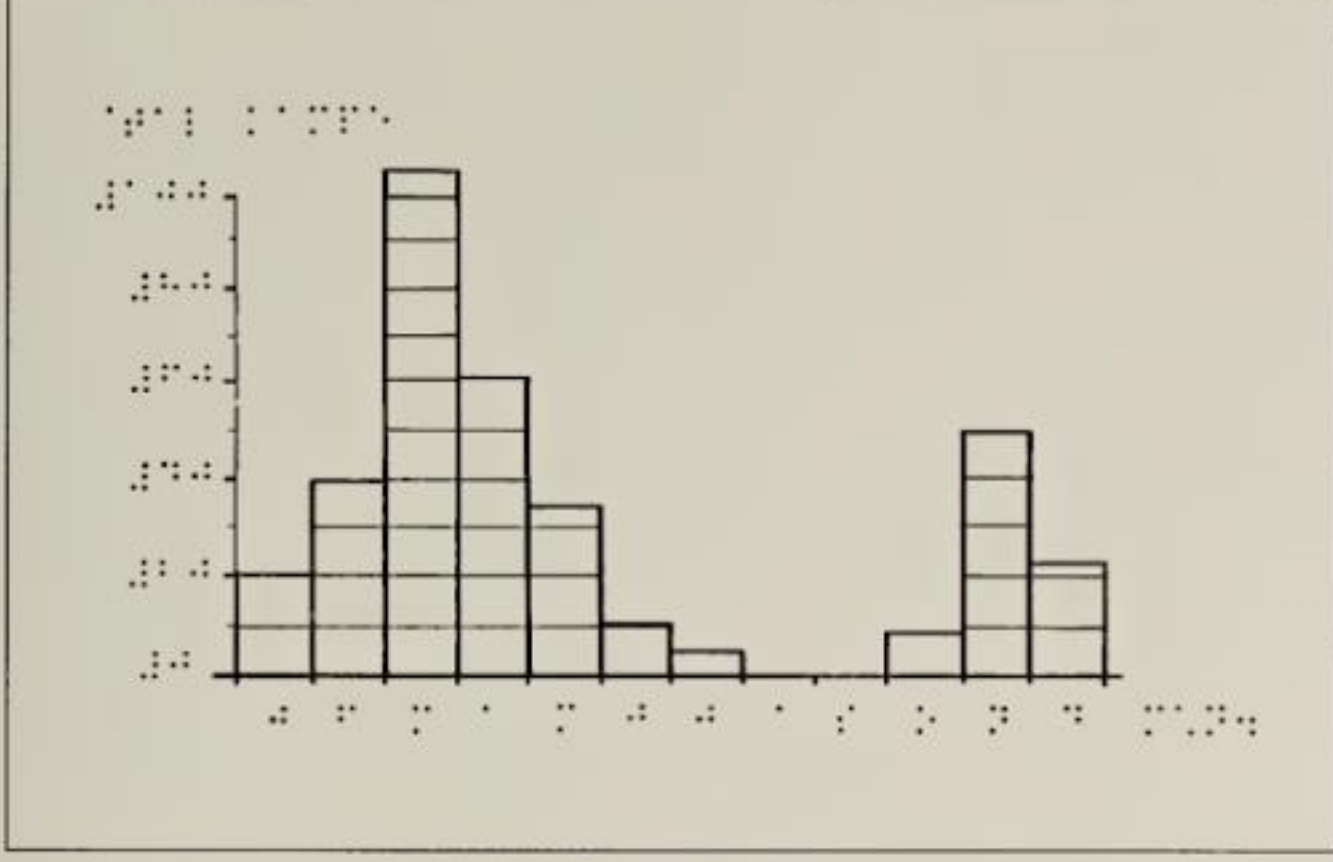
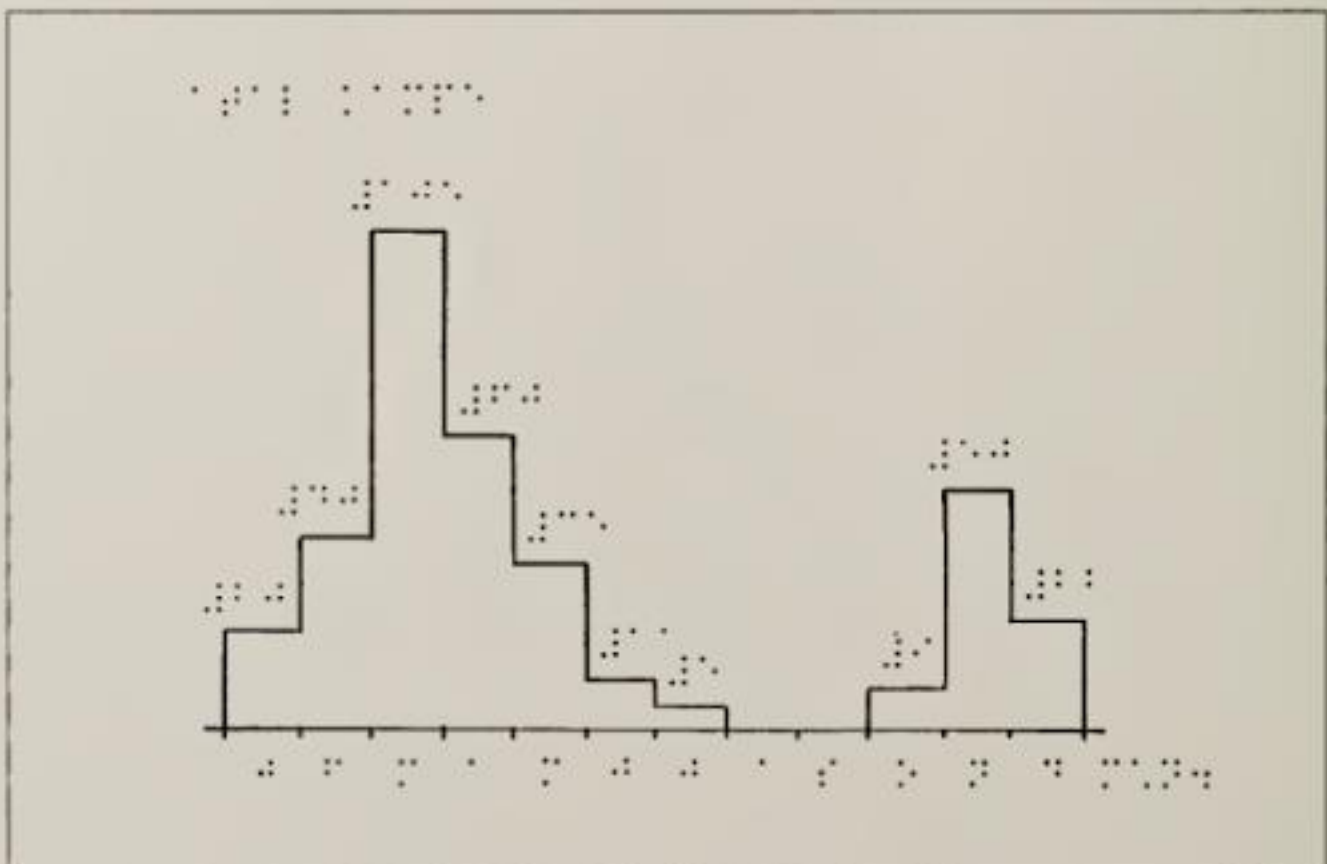
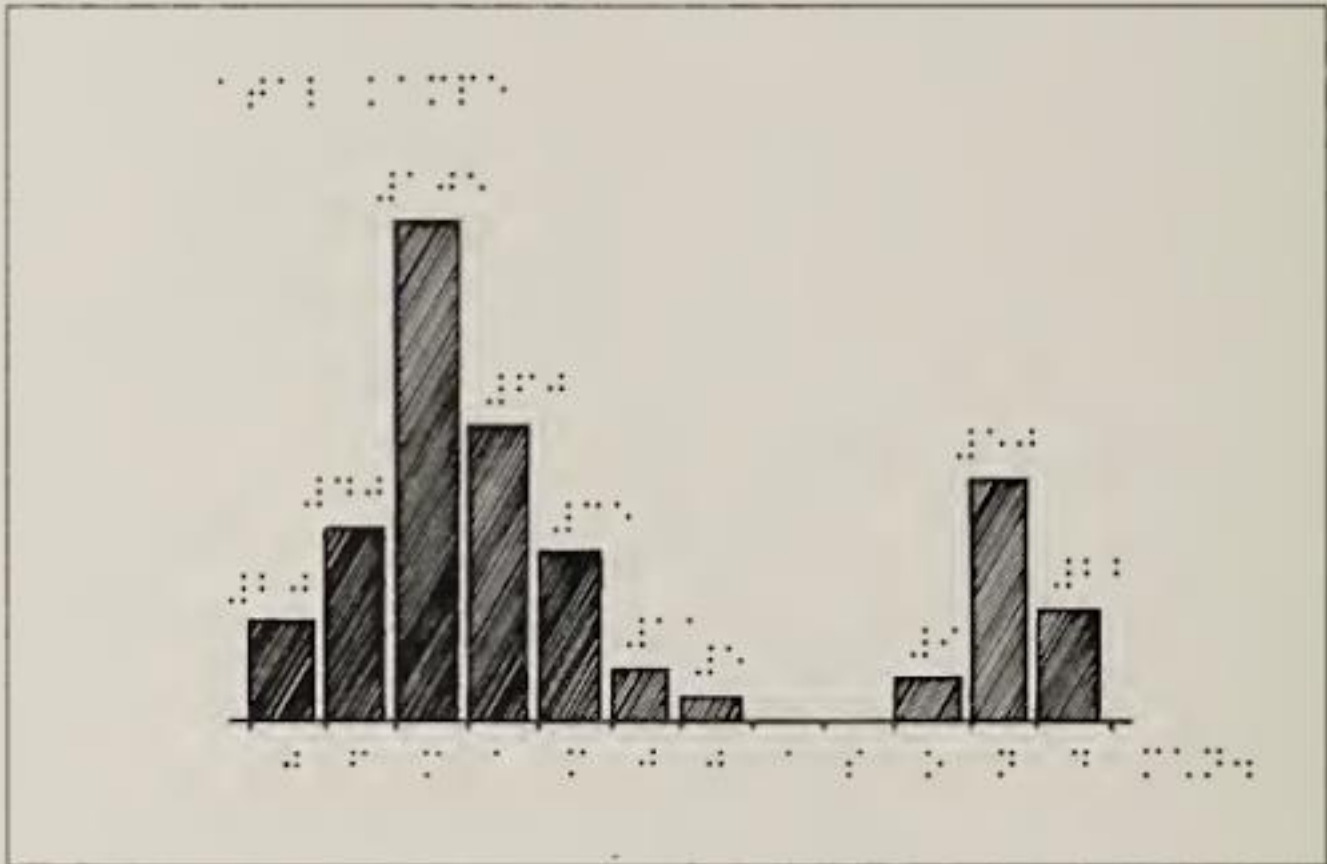
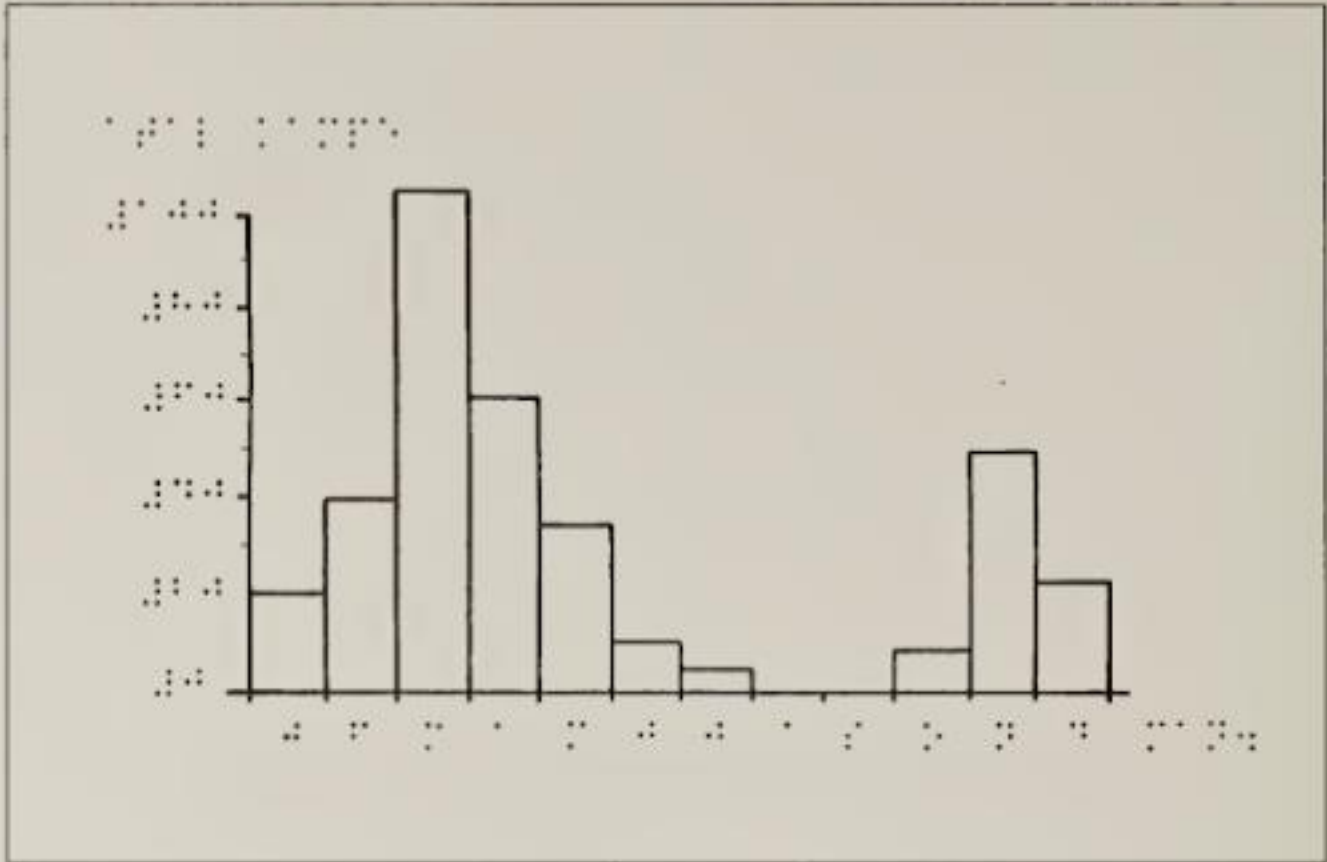
A braille table and an embossed graph used in the experiment are shown in Fig.1.

The Braille table contains a title at the top and two columns of data. The title is in Braille and appears to be '図表 1' (Figure Table 1). The data is organized into two columns, each with 10 rows of Braille characters.

図表 1	
1	2
3	4
5	6
7	8
9	10
11	12
13	14
15	16
17	18
19	20



A series of graphs based on the original graph.

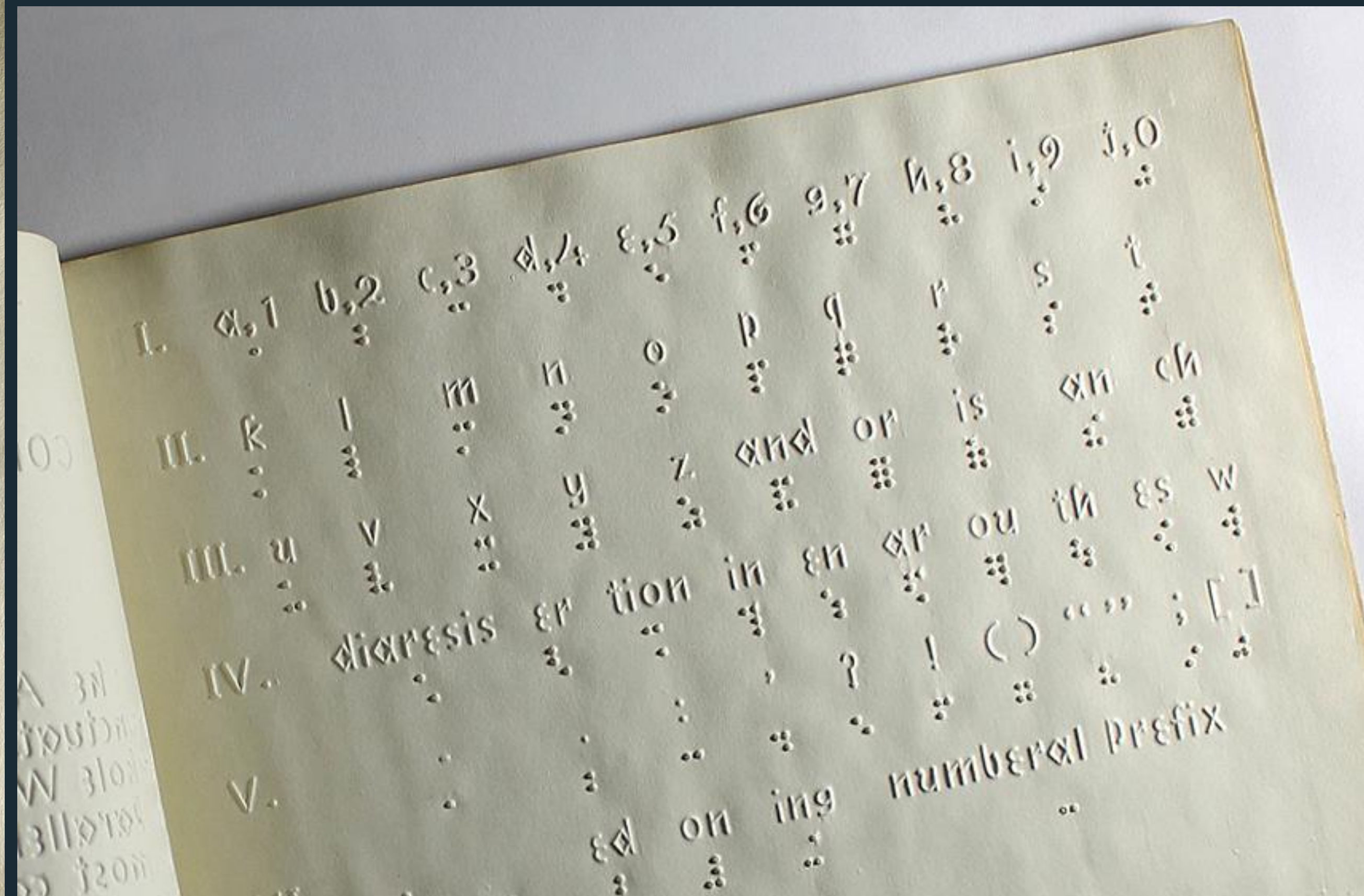


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 3/25/22



"They Aren't Built For Me": An Exploratory Study of Strategies for Measurement of Graphical Primitives in Tactile Graphics

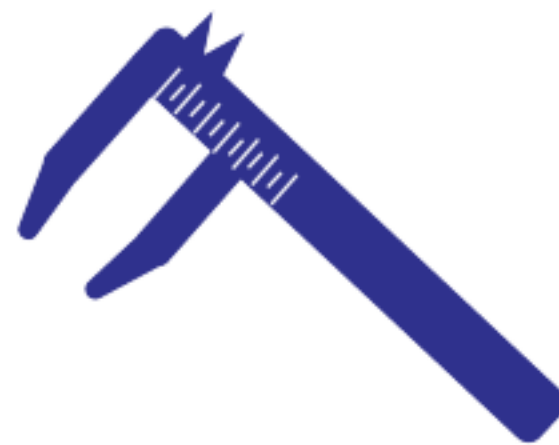
Areen Khalaila *
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Worcester Polytechnic Institute

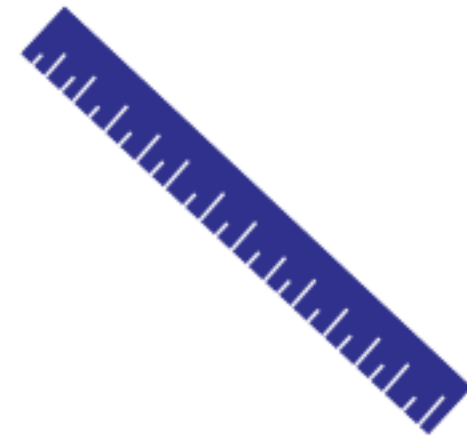
Nam Wook Kim [‡]
Boston College

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Brandeis University

In tactile perception, translatable strategies exist for some types of measurement such as length and angle...



(a) Caliper



(b) Ruler



(c) Bevel Gauge

... but not all measurements, such as area and popout.



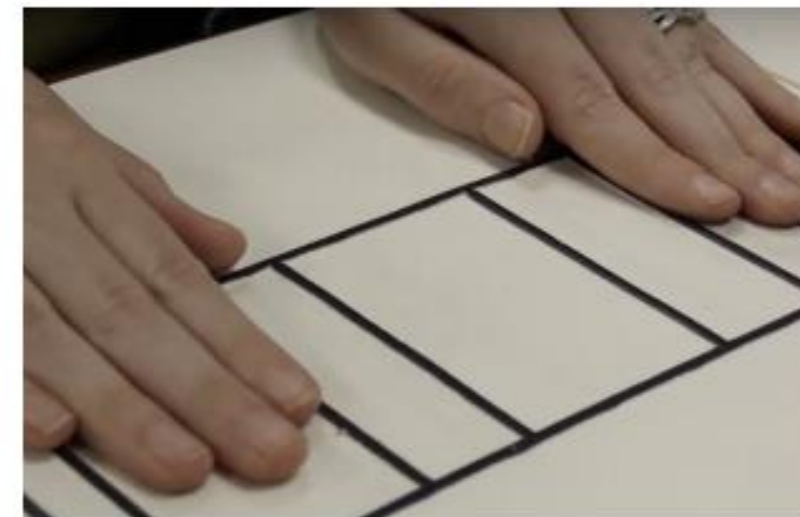
(d) Paint



(e) Stud Finder



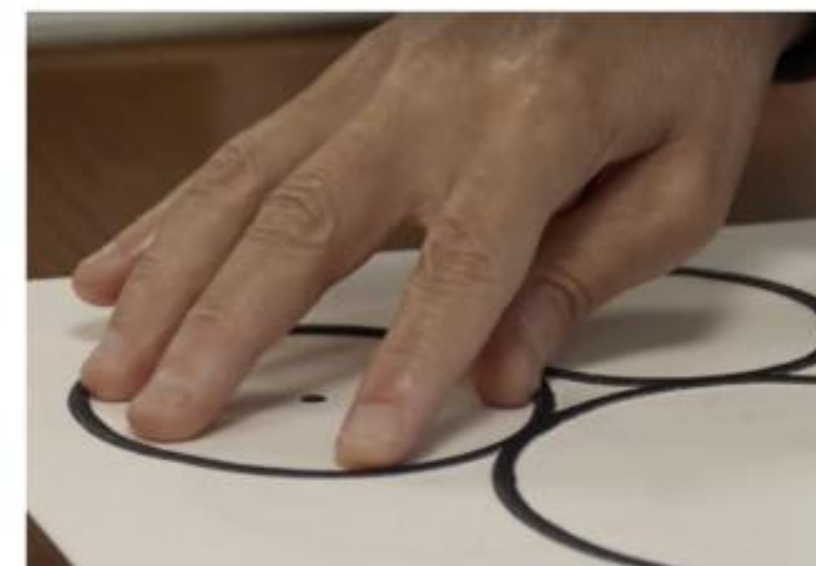
(a)



(b)



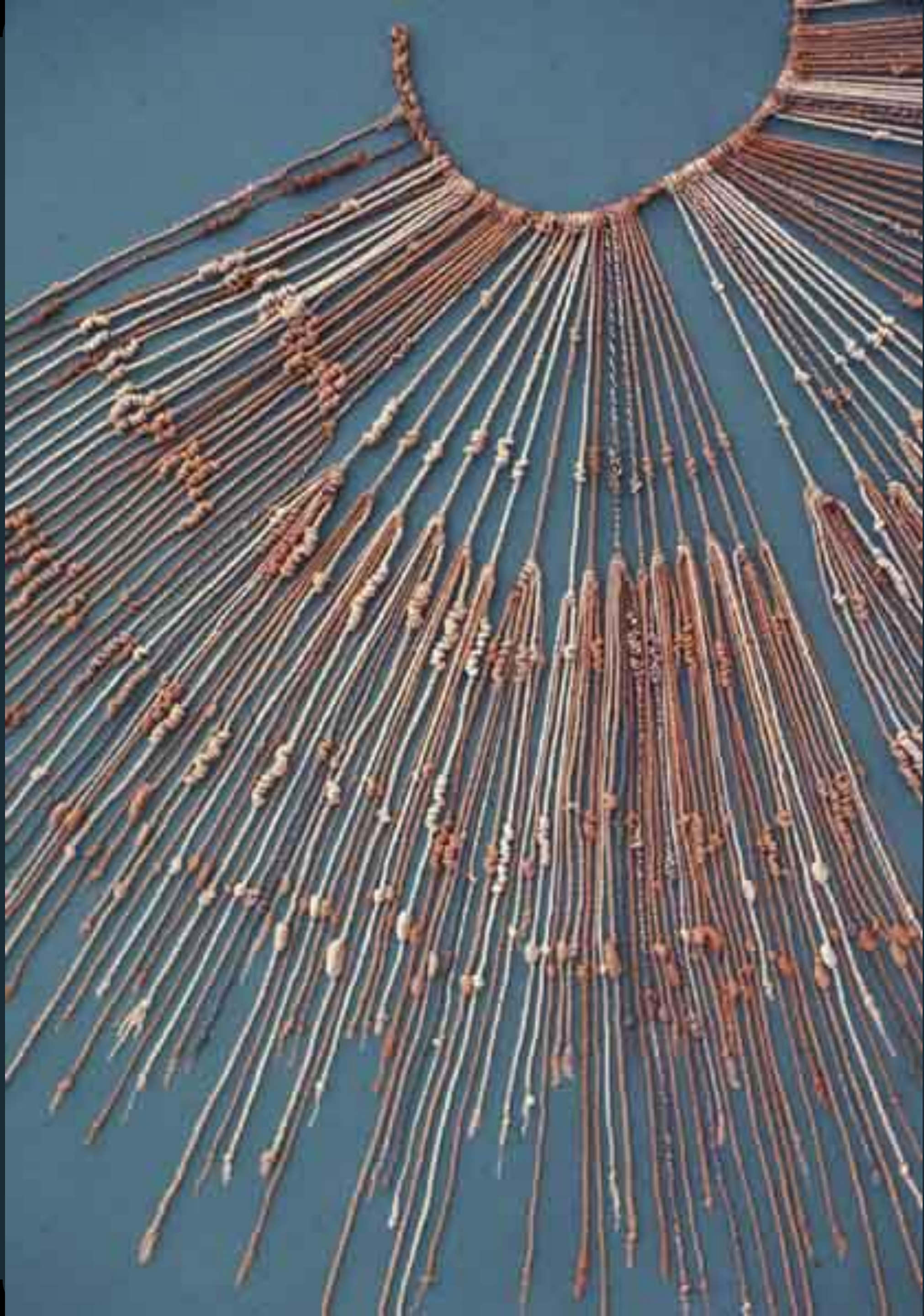
(c)



(d)



(e)



Quipus (c. 2600 BC)

Complex knotted ropes used in South America as data storage devices.

Played an important role in Inca administration.

From the *Gallery of Physical Visualization and Related Artifacts* by Pierre Dragicevic and Yvonne Jansen.



Polynesian Genealogical Instrument

Served as a memory aid for reciting genealogies during ceremonies.

Made of woven coconut fiber.

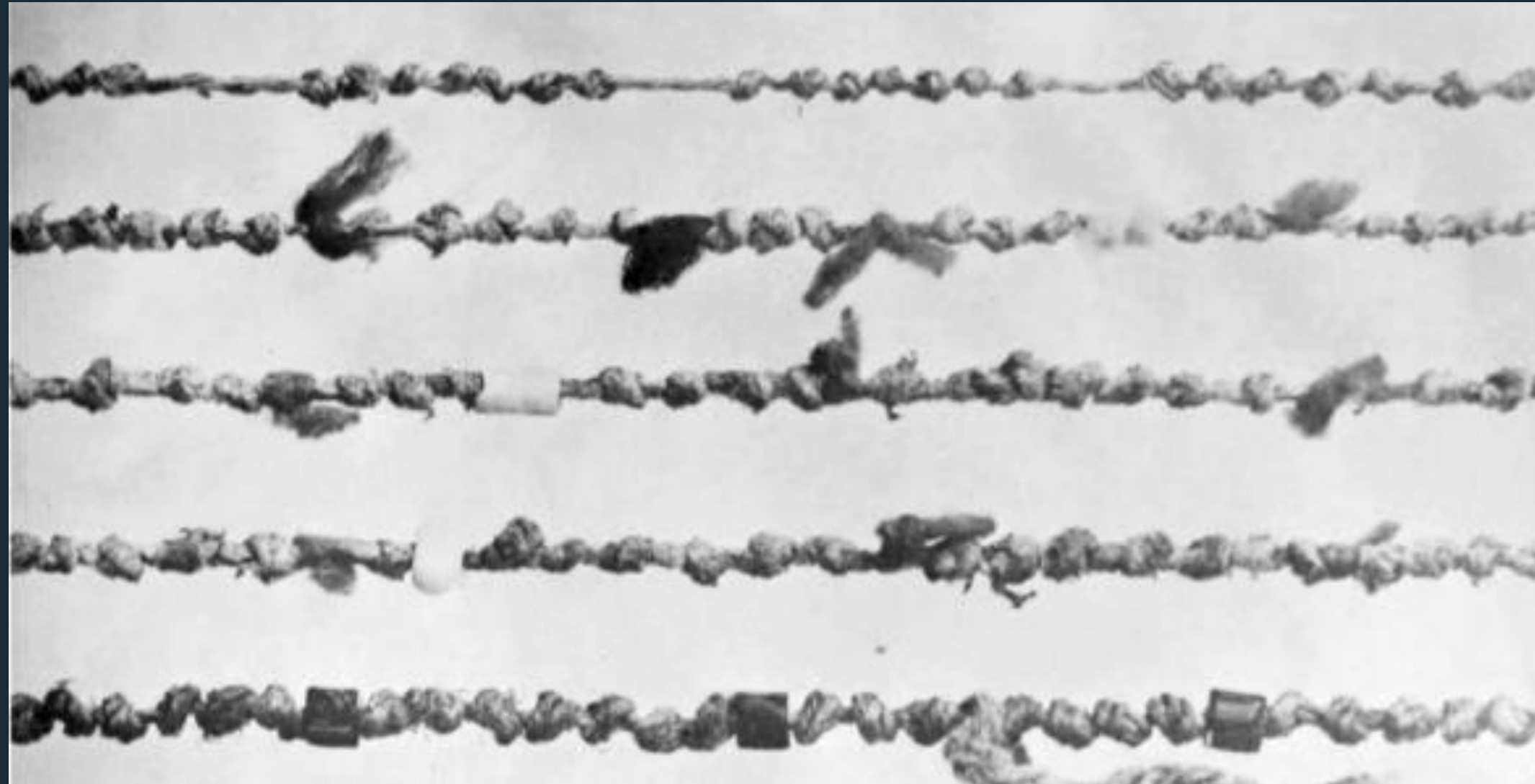
Each thread is a genealogical line, each knot is a generation.

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Yakama Time Ball

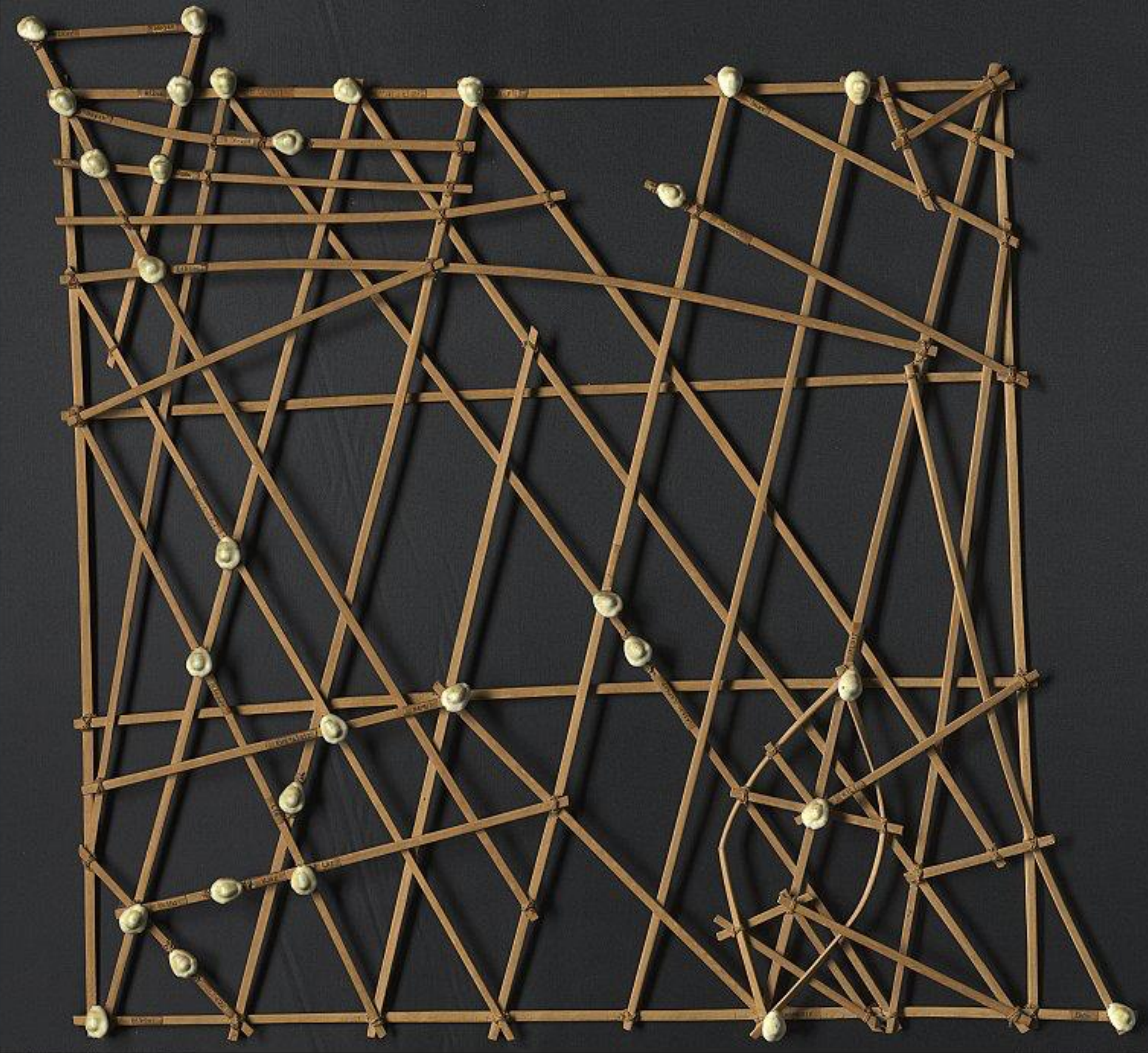
Women from the Yakama Native America tribe used strings of hemp as personal diaries.

Each major event in their life was represented by a knot, bead, or shell.



From the *Gallery of Physical Visualization and Related Artifacts* by Pierre Dragicevic and Yvonne Jansen.





Marshall Islands Stick Charts (c.1862)

Physical visualization show
ocean swell patterns.

Built by native
Micronesians from the
Marshall Islands to
facilitate canoe navigation.

From the *Gallery of Physical
Visualization and Related Artifacts*
by Pierre Dragicevic and Yvonne
Jansen.



DATA CUISINE

DATA CUISINE

WORKSHOP



Susanne Jaschko
Moritz Stefaner

+ local { organizers
chef
data
participants

Deconstructed Food Miles Smoothie

How far did the smoothie ingredients travel to get to Boston?



Liquid assets

Which parts of the population have access to liquid assets?

95.7%



US white

*Privateer Silver Serve
Rum*

82.6%



US black

*GrandTen Distilling
Medford*

84.8%



Caribbean Black

*Plantation 5-
Year Grand
Reserve*

63.3%



Dominican

Brugal Anejo Superior

57.2%



Puerto Rican

*Bacardi
Superior*



Sea Change

The effects of
climate change on
fish and shellfish
populations



Scents and Sensibility: Evaluating Information Olfactation

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